ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success



PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

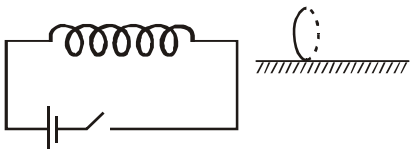
Electromagnetic Induction & Alternating Current

English Medium

EXERCISE (O-1)

A. Faraday law and Lenz law

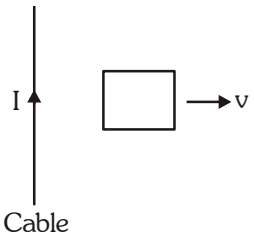
1. Figure shown a horizontal solenoid connected to a battery and a switch. A copper ring is place on a frictionless track, the axis of the ring being along the axis of the solenoid. As the switch is closed, the ring will



(A) remain stationary
(B) move towards the solenoid
(C) move away from the solenoid
(D) move towards the solenoid or away from it depending on which terminal (positive or negative) of the battery is connected to the left end of the solenoid.

EM0326

2. A rectangular coil is pulled away from the fixed current carrying cable as shown in the figure. The current induced in the loop is–



(A) Anti-clockwise
(B) Clockwise
(C) Zero
(D) May be clockwise or anticlockwise depends on speed of coil

EM0327

3. A magnet is moving towards a coil along its axis and the emf induced in the coil is ε. If the coil also starts moving towards the magnet with the same speed, the induced emf will be

(A) $\varepsilon/2$ (B) ε (C) 2ε (D) 4ε

EM0328

4. A cylindrical bar magnet is kept along the axis of a circular coil. If the magnet is rotated about its axis, then
(A) a current will be induced in the coil
(B) no current will be induced in the coil
(C) only an emf will be induced in the coil
(D) both a current and emf will be induced in the coil

EM0329

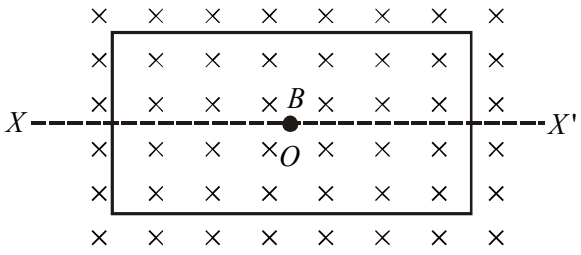
5. In a coil of area $10\ cm^2$ and 10 turns, the magnetic field is directed perpendicular to the plane and is changing at the rate of 10^8 gauss/second. The resistance of the coil is 20 ohm. The current in the coil will be

(A) 5 A (B) 0.5 A (C) 0.05 A (D) $5\times10^8\ A$

EM0330

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6. A rectangular coil is placed in a region having a uniform magnetic field B perpendicular to the plane of the coil. An emf will not be induced in the coil if the



(A) magnetic field is increased uniformly
(B) magnetic field is switched off
(C) coil is rotated about the axis XX'
(D) coil is rotated about an axis perpendicular to the plane of the coil and passing through its centre O

EM0331

7. A square of side *2* meters lies in the *x*-*y* plane in a region, where the magnetic field is given by $\vec{B} = B_0\left(2\hat{i} + 3\hat{j} + 4\hat{k}\right)T$, where B_0 is constant. The magnitude of flux passing through the square is:-

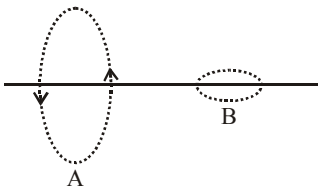
(A) 8 B_0 Wb. (B) 12 B_0 Wb. (C) 16 B_0 Wb. (D) $\sqrt{4\times 29}B_0 Wb$

EM0049

8. **Statement-1 :** When a magnet is made to fall freely through a closed coil, its acceleration is always less than acceleration due to gravity.
and
Statement-2 : Current induced in the coil opposes the motion of the magnet, as per Lenz's law.
(A) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is true, Statement-2 is true, Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

EM0050

9. In the given figure the centre of a small conducting circular loop B lies on the axis of bigger circular loop A and their axis are mutually perpendicular. An anticlockwise (when viewed from the side of B) current in the loop A start increasing then :-



(A) current induced in the loop B is in clockwise direction (when viewed from above the B)
(B) current induced in the loop B is in anti-clockwise direction (when viewed from above the B)
(C) current must induced in the loop B but its direction can not be predicted
(D) no current is induced in the loop B

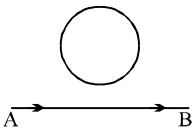
EM0051

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10. Two identical coaxial circular loops carry a current *i* each circulating in the same direction. If the loops approach each other
(A) the current in each will decrease
(B) the current in each will increase
(C) the current in each will remain the same
(D) the current in one will increase and in other will decrease

EM0053

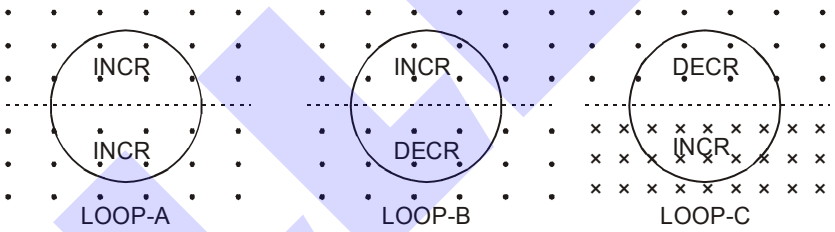
11. In the arrangement shown in given figure current from A to B is increasing in magnitude. Induced current in the loop will



(A) have clockwise direction
(B) have anticlockwise direction
(C) be zero
(D) oscillate between clockwise and anticlockwise

EM0054

12. Three identical conducting circular loops are placed in uniform magnetic fields. Inside each loop, there are two magnetic field regions, separated by dashed line that coincides with a diameter, as shown. Magnetic fields may either be increasing (marked as INCR) or decreasing (marked as DECR) in magnitude at the same rates. If I_A, I_B and I_C are the magnitudes of the induced currents in the loops A, B and C respectively then choose the **CORRECT** relation :-

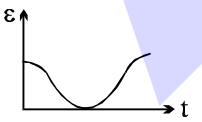


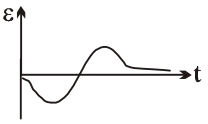
(A) $I_A > I_B = I_C$ (B) $I_A = I_C > I_B$ (C) $I_A = I_B = I_C$ (D) $I_C > I_A > I_B$

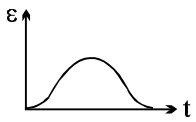
EM0055

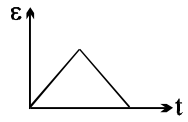
13. The variation of induced emf (ε) with time (t) in a coil if a short bar magnet is moved along its axis with a constant velocity is best represented as **[JEE 2004(Scr)]**



(A) 

(B) 

(C) 

(D) 

EM0058

14. A rectangular coil of single turn, having area A, rotates in a uniform magnetic field B with an angular velocity ω about an axis perpendicular to the field. If initially the plane of coil is perpendicular to the field, then the average induced e.m.f. when it has rotated through 90° is :-

(A) $\frac{\omega BA}{\pi}$ (B) $\frac{\omega BA}{2\pi}$ (C) $\frac{\omega BA}{4\pi}$ (D) $\frac{2\omega BA}{\pi}$

EM0157

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B. Motional EMF

15. A square loop of side a and resistance R is moved in the region of uniform magnetic field B (loop remaining completely inside field), with a velocity v through a distance x . The work done is :

(A) $\frac{B\ell^2vx}{R}$ (B) $\frac{2B^2\ell^2vx}{R}$ (C) $\frac{4B^2\ell^2vx}{R}$ (D) 0

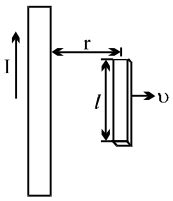
EM0062

16. The radius of a coil decreases steadily at the rate of 10^{-2} m/s. A constant and uniform magnetic field of induction 10^{-3} Wb/m^2 acts perpendicular to the plane of the coil. The radius of the coil when the induced e.m.f. in the coil is 1μV, is :-

(A) $\frac{2}{\pi}$ cm (B) $\frac{3}{\pi}$ cm (C) $\frac{4}{\pi}$ cm (D) $\frac{5}{\pi}$ cm

EM0146

17. A conducting rod moves with constant velocity υ perpendicular to the long, straight wire carrying a current I as shown. Compute the emf generated between the ends of the rod.



(A) $\frac{\mu_0 \upsilon I l}{\pi r}$ (B) $\frac{\mu_0 \upsilon I l}{2\pi r}$ (C) $\frac{2\mu_0 \upsilon I l}{\pi r}$ (D) $\frac{\mu_0 \upsilon I l}{4\pi r}$

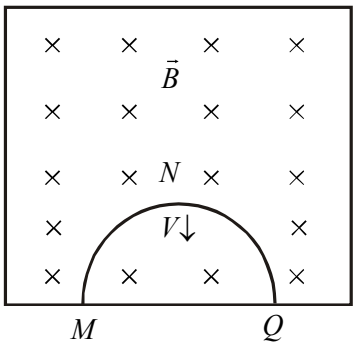
EM0151

18. A rod a length l rotates with a small but uniform angular velocity ω about its perpendicular bisector. A uniform magnetic field B exists parallel to the axis of rotation. The potential difference between the centre of the rod and an end is

(A) zero (B) $\frac{1}{8}\omega Bl^2$ (C) $\frac{1}{2}\omega Bl^2$ (D) $B\omega l^2$

EM0332

19. A thin semicircular conducting ring of radius R is falling with its plane vertical in a horizontal magnetic induction $\vec{B}$. At the position MNQ, the speed of the ring is V and the potential difference developed across the ring is



(A) zero (B) $BV\pi R^2/2$ and M is at a higher potential

(C) πRBV and Q is at a higher potential (D) $2RBV$ and Q is at a higher potential

EM0333

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20. Two identical cycle wheels (geometrically) have different number of spokes connected from centre to rim. One is having 20 spokes and other having only 10 (the rim and the spokes are resistanceless). One resistance of value R is connected between centre and rim. The current in R will be:

(A) double in first wheel than in the second wheel

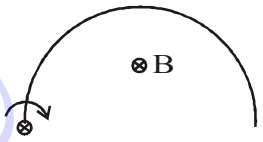
(B) four times in first wheel than in the second wheel

(C) will be double in second wheel than that of the first wheel

(D) will be equal in both these wheels.

EM0334

21. A semicircular wire of radius R is rotated with constant angular velocity ω about an axis passing through one end and perpendicular to the plane of the wire. There is a uniform magnetic field of strength B. The induced e.m.f. between the ends is–

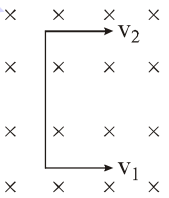


(A) $B\omega R^2/2$ (B) $2B\omega R^2$

(C) is variable (D) none of these

EM0335

22. In the figure magnetic field points into the plane of paper and the rod of length l is moving in this field such that the bottom most point has a velocity v_1 and the topmost point has the velocity v_2 $(v_2 > v_1)$. the emf induced is given by :

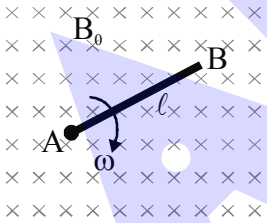


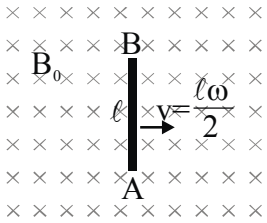
(A) Bv_1l (B) Bv_2l

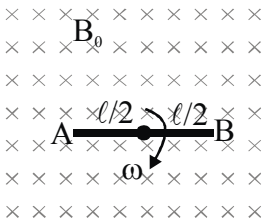
(C) $\frac{1}{2}B(v_2+v_1)\,l$ (D) $\frac{1}{2}B(v_2-v_1)\,l$

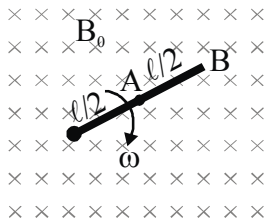
EM0336

23. The figure shows a rod of length ℓ with points A and B on it. The rod is moved in a uniform magnetic field (B_0) in different ways as shown. In which case potential difference $(V_A - V_B)$ between A & B is minimum?

(A) 

(B) 

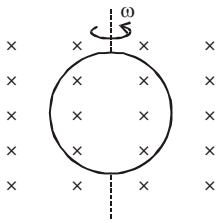
(C) 

(D) 

EM0337

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24. The e.m.f. induced in a coil of wire, which is rotating in a magnetic field, does not depend on



(A) the angular speed of rotation
(B) the area of the coil
(C) the number of turns on the coil
(D) the resistance of the coil

EM0067

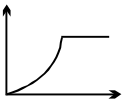
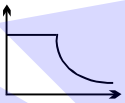
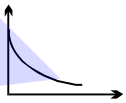
25. A ring of resistance 10Ω, radius 10cm and 100 turns is rotated at a rate 100 revolutions per second about its diameter is perpendicular to a uniform magnetic field of induction 10mT. The amplitude of the current in the loop will be nearly (Take : $\pi^2 = 10$)

(A) 200A (B) 2A (C) 0.002A (D) none of these

EM0068

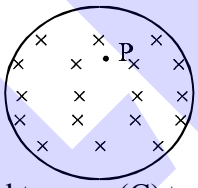
C. Induced electric field

26. A uniform but time variant magnetic field exists in a cylindrical region directed along the axis of cylinder of radius R. The graph of induced electric field at a given time v/s. r is (r = distance from axis)

(A)  (B)  (C)  (D) 

EM0064

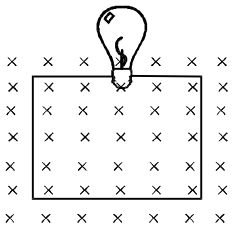
27. Figure shows a uniform magnetic field B confined to a cylindrical volume and is increasing at a constant rate. The instantaneous acceleration experienced by an electron placed at P is



(A) zero (B) towards right (C) towards left (D) upwards

EM0070

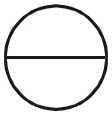
28. A square wire loop of 10.0 cm side lies at right angles to a uniform magnetic field of 20T. A 10 V light bulb is in a series with the loop as shown in the fig. The magnetic field is decreasing steadily to zero over a time interval Δt. The bulb will shine with full brightness if Δt is equal to :-



(A) 20 ms (B) 0.02 ms
(C) 2 ms (D) 0.2 ms

EM0144

29. A wire loop consists of two parts as shown below. It is placed in a magnetic field directed into the plane of the paper and increases with time. Which of the following shows the induced current correctly?

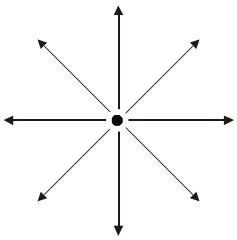
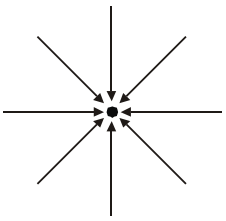
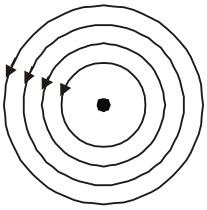
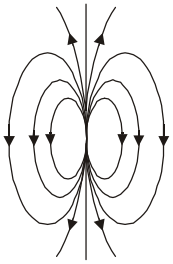


(A)  (B)  (C)  (D) 

EM0338

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30. Which of the field patterns given below is valid for electric field as well as for magnetic field?
[JEE 2011]

(A)  (B)  (C)  (D) 

EM0197

D. Self inductance and mutual inductance

31. L, C and R represent physical quantities inductance, capacitance and resistance. The combination which has the dimensions of frequency is

(A) $\frac{1}{RC}$ and $\frac{R}{L}$ (B) $\frac{1}{\sqrt{RC}}$ and $\sqrt{\frac{R}{L}}$ (C) $\sqrt{LC}$ (D) $\frac{C}{L}$

EM0085

32. A coil of $N = 100$ turns carries a current $I = 5\ A$ and creates a magnetic flux $\phi = 10^{-5}\ Tm^2$ per turn. The value of its inductance L will be
(A) 0.05 mH (B) 0.10 mH (C) 0.15 mH (D) 0.20 mH

EM0339

33. When the number of turns in a coil is doubled without any change in the length and area of the coil, its self inductance becomes
(A) Four times (B) Doubled (C) Halved (D) Unchanged

EM0340

34. Two coils are placed close to each other. The mutual inductance of the pair of coils depends upon-
(A) the rates at which currents are changing in the two coils **[AIEEE - 2003]**
(B) relative position and orientation of the two coils
(C) the materials of the wires of the coils
(D) the currents in the two coils

EM0083

35. The mutual inductance between two coils is 1.25 *henry*. If the current in the primary changes at the rate of 80 *ampere/second*, then the induced e.m.f. in the secondary is
(A) 12.5 *V* (B) 64.0 *V* (C) 0.016 *V* (D) 100.0 *V*

EM0341

36. Two coils A and B have coefficient of mutual inductance M = 2H. The magnetic flux passing through coil A changes by 4 Weber in 10 seconds due to the change in current in B. Then
(A) the change in current in B in this time interval is 0.5 A
(B) the change in current in B in this time interval is 2A
(C) the change in current in B in this time interval is 8A
(D) a change in current of 1A in coil A will produce a change in flux passing through B by 4 Weber.

EM0171

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37. For perfect coupling of two coils of inductance L_1 and L_2, their mutual inductance M should be given by

(A) $M = L_1 L_2$ (B) $M = \frac{L_1}{L_2}$ (C) $M = \sqrt{L_1 L_2}$ (D) $M = \left[\frac{L_1}{L_2}\right]^{1/2}$

EM0342

38. A long straight wire is placed along the axis of a circular ring of radius R. The mutual inductance of this system is :-

(A) $\frac{\mu_0 R}{2}$ (B) $\frac{\mu_0 \pi R}{2}$ (C) $\frac{\mu_0}{2}$ (D) 0

EM0082

39. A small coil of radius r is placed at the centre of a large coil of radius R, where R >> r. The coils are coplanar. The coefficient of mutual inductance between the coils is :-

(A) $\frac{\mu_0 \pi r}{2R}$ (B) $\frac{\mu_0 \pi r^2}{2R}$ (C) $\frac{\mu_0 \pi r^2}{2R^2}$ (D) $\frac{\mu_0 \pi r}{2R^2}$

EM0081

40. A small square loop of wire of side l is placed inside a large square loop of wire of side L($L >> l$). The loop are coplanar & their centres coincide. The mutual inductance of the system is proportional to :

(A) $\frac{\ell}{L}$ (B) $\frac{\ell^2}{L}$ (C) $\frac{L}{\ell}$ (D) $\frac{L^2}{\ell}$

EM0084

41. Two coaxial solenoids are made by winding thin insulated wire over a pipe of cross-sectional area A = 10 cm^2 and length = 20 cm. If one of the solenoids has 300 turns and the other 400 turns, their mutual inductance is $\left(\mu_0 = 4\pi \times 10^{-7} \text{TmA}^{-1}\right)$ **[AIEEE - 2008]**

(A) 2.4 $\pi \times 10^{-5}$ H (B) 4.8 $\pi \times 10^{-4}$H (C) 4.8 $\pi \times 10^{-5}$H (D) 2.4 $\pi \times 10^{-4}$H

EM0343

42. If the difference between the equivalent inductance in the following figures is nL then find the value of n.

Given coupling coefficient between two inductance is $C = \frac{1}{\sqrt{2}}.7$

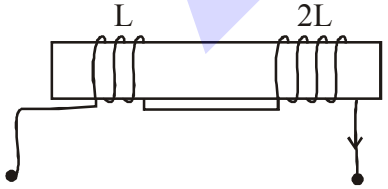


Figure (A)

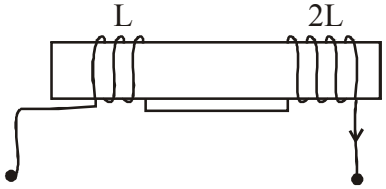


Figure (B)

(A) 2 (B) 3 (C) 4 (D) 5

EM0112

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E. R-L circuit

43. An LR circuit with a battery is connected at t =0. Which of the following quantities is not zero just after the connection ?

(A) current in the circuit
(B) magnetic field energy in the inductor
(C) power delivered by the battery
(D) emf induced in the inductor

EM0344

44. An ideal coil of 10 H is joined in series with a resistance of 5 ohm and a battery of 5 V. 2 seconds after joining, the current flowing in the circuit, in amperes, will be

(A) e^{-1} (B) $1-e^{-1}$ (C) $1-e$ (D) e

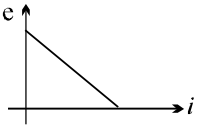
EM0345

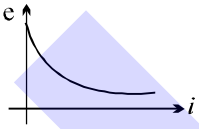
45. An emf of 15 V is applied in a circuit containing 5 H inductance and 10Ω resistance. The ratio of the currents at time $t=\infty$ and at t = 1 is

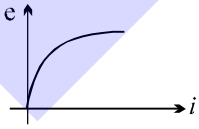
(A) $\frac{e^{1/2}}{e^{1/2}-1}$ (B) $\frac{e^2}{e^2-1}$ (C) $1-e^{-1}$ (D) e^{-1}

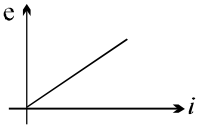
EM0346

46. In an L-R circuit connected to a battery of constant e.m.f. E switch S is closed at time t = 0. If e denotes the magnitude of induced e.m.f. across inductor and *i* the current in the circuit at any time t. Then which of the following graphs shows the variation of e with *i* ?

(A) 

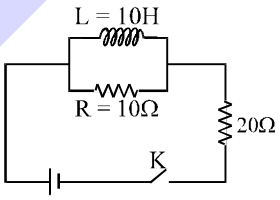
(B) 

(C) 

(D) 

EM0072

47. Two resistors of 10Ω and 20Ω and an ideal inductor of 10H are connected to a 2V battery as shown. The key K is shorted at time t = 0. Find the initial (t = 0) and final ($t \to \infty$) currents through battery.



(A) $\frac{1}{15}A, \frac{1}{10}A$ (B) $\frac{1}{10}A, \frac{1}{15}A$ (C) $\frac{2}{15}A, \frac{1}{10}A$ (D) $\frac{1}{15}A, \frac{2}{25}A$

EM0079

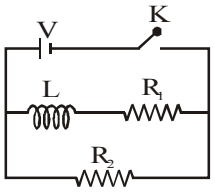
48. A coil of inductance 5H is joined to a cell of emf 6V through a resistance 10Ω at time t = 0. The emf across the coil at time t = *l*n $\sqrt{2}$ s is :

(A) 3V (B) 1.5 V (C) 0.75 V (D) 4.5 V

EM0087

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49. In the circuit show below, the key K is closed at t = 0. The current through the battery is : **[AIEEE - 2010]**

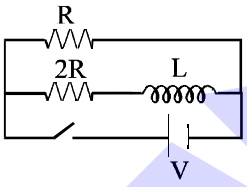


(A) $\frac{V(R_1+R_2)}{R_1R_2}$ at t = 0 and $\frac{V}{R_2}$ at t = ∞ (B) $\frac{VR_1R_2}{\sqrt{R_1^2+R_2^2}}$ at t = 0 and $\frac{V}{R_2}$ at t = ∞

(C) $\frac{V}{R_2}$ at t = 0 and $\frac{V(R_1+R_2)}{R_1R_2}$ at t = ∞ (D) $\frac{V}{R_2}$ at t = 0 and $\frac{VR_1R_2}{\sqrt{R_1^2+R_2^2}}$ at t = ∞

EM0175

50. Find the ratio of time constant in build up and decay in the circuit as shown in figure :-



(A) 1 : 1 (B) 3 : 2 (C) 2 : 3 (D) 1 : 3

EM0090

51. In a L–R decay circuit, the initial current at t = 0 is I. The total charge that has flown through the resistor till the energy in the inductor has reduced to one–fourth its initial value, is

(A) LI/R (B) $LI/2R$ (C) $LI\sqrt{2}/R$ (D) None

EM0092

52. A current of 2A is increasing at a rate of 4 A/s through a coil of inductance 2H. The energy stored in the inductor per unit time is :-

(A) 2 J/s (B) 1 J/s (C) 16 J/s (D) 4 J/s

EM0073

53. An inductor coil stores U energy when i current is passed through it and dissipates energy at the rate of P. The time constant of the circuit, when this coil is connected across a battery of zero internal resistance is :-

(A) $\frac{4U}{P}$ (B) $\frac{U}{P}$ (C) $\frac{2U}{P}$ (D) $\frac{2P}{U}$

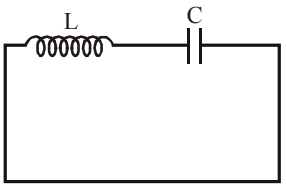
EM0080

F. L-C oscillation

54. In an LC circuit the capacitor has maximum charge q_0. The value of $\left(\frac{dI}{dt}\right)_{max}$ is :-

(A) $\frac{q_0}{LC}$ (B) $\frac{q_0}{\sqrt{LC}}$

(C) $\frac{q_0}{2LC}$ (D) $\frac{2q_0}{LC}$



EM0095

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55. The inductor in a L–C oscillation has a maximum potential difference of 16 V and maximum energy of 640 μJ. Find the value of capacitor in μF in L–C circuit.

(A) 5 (B) 4 (C) 3 (D) 2

EM0093

56. A condenser of capacity 6 μF is fully charged using a 6-volt battery. The battery is removed and a resistanceless 0.2 mH inductor is connected across the condenser. The current which is flowing through the inductor when one-third of the total energy is in the magnetic field of the inductor is :-

(A) 0.1 A (B) 0.2 A (C) 0.4 A (D) 0.6 A

EM0094

G. AC circuits

57. If R, C and L be the resistance, capacitance and inductance in a circuit in which ac of frequency f is set up, then which of the following has the dimensions of R ?

(A) fC (B) fL (C) C/f (D) L/f

EM0347

58. An alternating current is given by the equation $i = i_1 \cos\omega t + i_2 \sin\omega t$. The *r.m.s.* current is given by

(A) $\frac{1}{\sqrt{2}}(i_1 + i_2)$ (B) $\frac{1}{\sqrt{2}}(i_i + i_2)^2$ (C) $\frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$ (D) $\frac{1}{2}(i_1^2 + i_2^2)^{1/2}$

EM0348

59. If I_1, I_2, I_3 and I_4 are the respective r.m.s. values of the time varying currents as shown in the four cases I, II, III and IV. Then identify the correct relations.



(A) $I_1 = I_2 = I_3 = I_4$ (B) $I_3 > I_1 = I_2 > I_4$ (C) $I_3 > I_4 > I_2 = I_1$ (D) $I_3 > I_2 > I_1 > I_4$

EM0097

60. Voltage and current in an ac circuit are given by $V = 5\sin\left(100\pi t - \frac{\pi}{6}\right)$ and $I = 4\sin\left(100\pi t + \frac{\pi}{6}\right)$

(A) Voltage leads the current by 30° (B) Current leads the voltage by 30°

(C) Current leads the voltage by 60° (D) Voltage leads the current by 60°

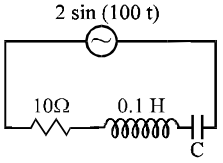
EM0349

61. An current is given by $I = I_0 + I_1 \sin\omega t$ then its rms value will be

(A) $\sqrt{I_0^2 + 0.5I_1^2}$ (B) $\sqrt{I_0^2 + 0.5I_0^2}$ (C) 0 (D) $I_0/\sqrt{2}$

EM0163

62. The power factor of the circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to

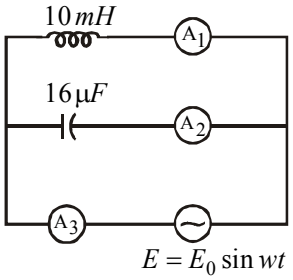


(A) 400 μF (B) 300 μF (C) 500 μF (D) 200 μF

EM0098

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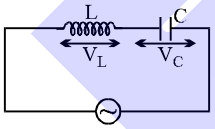
63. An inductor of 10 mH and a capacitor of $16\mu F$ are connected in the circuit as shown in the fig. The frequency of the power supply is equal to the resonant frequency of the circuit. Which ammeter will read zero ampere ?

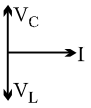


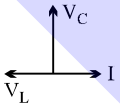
(A) A_1 (B) A_2 (C) A_3 (D) none

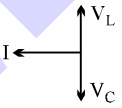
EM0350

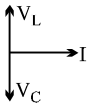
64. The current I, potential difference V_L across the inductor and potential difference V_C across the capacitor in circuit as shown in the figure are best represented vectorially as



(A) 

(B) 

(C) 

(D) 

EM0105

65. In a series R-L-C circuit, the frequency of the source is half of the resonance frequency. The nature of the circuit will be

(A) capacitive (B) inductive

(C) purely resistive (D) data insufficient

EM0106

66. When a resistance R is connected in series with an element A, the electric current is found to be lagging behind the voltage by angle θ_1. When the same resistance is connected in series with element B, current leads voltage by θ_2. When R, A, B are connected in series, the current now leads voltage by θ. Assume same AC source is used in all cases, then

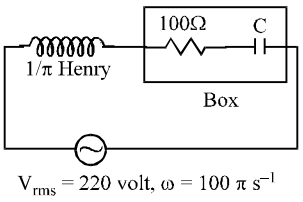
(A) $\theta = \theta_2 - \theta_1$ (B) $\tan\theta = \tan\theta_2 - \tan\theta_1$

(C) $\theta = \dfrac{\theta_1 + \theta_2}{2}$ (D) None of these

EM0162

node06\B0BC-BD\Kota\JEE\Advanced\Enthuse\Phy\Module\3-Magnetic Effects of Current\EM\8.AC\Eng\02_EM\8.AC\02_Exercises.p65

67. In the circuit, as shown in the figure, if the value of R.M.S current is 2.2 ampere, the power factor of the box is



(A) $\frac{1}{\sqrt{2}}$ (B) 1 (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

EM0099

68. In series LR circuit $X_L = 3R$. Now a capacitor with $X_C = R$ is added in series. Ratio of new to old power factor is

(A) 1 (B) 2 (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

EM0104

69. The phase difference between voltage and current during resonance in an LCR series circuit is

(A) zero (B) $\pi/4$ (C) $\pi/2$ (D) π

EM0351

70. The power in ac circuit is given by $P = E_{rms} I_{rms} \cos\phi$. The value of cos ϕ in series LCR circuit at resonance is:

(A) zero (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$

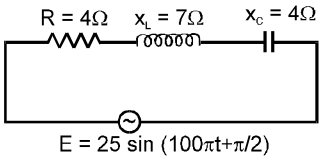
EM0100

71. When a voltage $v_s = 200\sqrt{2}$ sin (ωt + 15°) is applied to an AC circuit the current in the circuit is found to be i = 2 sin (ωt + π/4) then average power consumed in the circuit is

(A) 200 watt (B) $400\sqrt{2}$ watt (C) $100\sqrt{6}$ watt (D) $200\sqrt{2}$ watt

EM0352

72. In the series LCR circuit as shown in figure, the heat developed in 80 seconds and amplitude of wattless current is :



(A) 4000 J, 3A (B) 8000 J, 3A (C) 4000 J, 4A (D) 8000 J, 5A

EM0353

73. In ac circuit when ac ammeter is connected it reads i current. If a student uses dc ammeter in place of ac ammeter the reading in the dc ammeter will be:

(A) $\frac{i}{\sqrt{2}}$ (B) $\sqrt{2}$ i (C) 0.637 i (D) zero

EM0101

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74. A square wave current switching rapidly between + 0.5 ampere and –0.5 ampere is passed through an A.C ammeter. Then the reading shown by it, is

(A) 0.25 ampere (B) 0.5 ampere (C) $\frac{0.5}{\sqrt{2}}$ ampere (D) $0.5\times\sqrt{2}$ ampere

EM0354

75. Which of the following helps in the operation of the choke coil ?

(A) self induction (B) Mutual inductance (C) Eddy currents (D) none of the above

EM0355

76. An ac emf of $V = 200\sin(100\pi t)$ volt is connected to a choke of negligible resistance. In order to produce current of amplitude 1A, the inductance of choke should be

(A) 200 H (B) $\frac{2}{\pi}H$ (C) $\frac{1}{\pi}H$ (D) $\frac{\sqrt{2}}{\pi}H$

EM0356

77. An ideal inductor takes a current of 10A when connected to a 125 V, 50 Hz ac supply. A pure resistor across the same source takes 12.5 A. If the two are connected in series across a $100\sqrt{2}V$, 40 Hz supply, the current through the circuit will be

(A) 10 A (B) 12.5 A (C) 20 A (D) 25 A

EM0357

78. In a transformer, number of turns in the primary are 140 and that in the secondary are 280. If current in primary is 4A, then that in the secondary is

(A) 4A (B) 2A (C) 6A (D) 10 A

EM0108

79. An ideal efficient transformer has a primary power input of 10kW. The secondary current when the transformer is on load is 25A. If the primary : secondary turns ratio is 8 : 1, then the potential difference applied to the primary coil is

(A) $\frac{10^4\times8^2}{25}$V (B) $\frac{10^4\times8}{25}$V (C) $\frac{10^4}{25\times8}$V (D) $\frac{10^4}{25\times8^2}$V

EM0110

80. The core of any transformer is laminated so as to -

(A) Make it light weight (B) Make it robust and strong

(C) Increase the secondary voltage (D) Reduce the energy loss due to eddy current

EM0111

H. Damped and Forced oscillation

81. An RLC circuit has a capacitance of 12μF, an inductance of 25mH, and a resistance of 60Ω. The current oscillates with an angular frequency of :

(A) 1.2×10^3 rad/s (B) 1.4×10^3 rad/s (C) 1.8×10^3 rad/s (D) 2.2×10^3 rad/s

(E) 2.6×10^3 rad/s

EM0114

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82. An RLC circuit has an inductance of 25mH and a capacitance of 5.0μF. The charge on the capacitor does NOT oscillate but rather decays exponentially to zero. The resistance in the circuit must be:

(A) greater than or equal to $100\sqrt{2}\Omega$

(B) less than $100\sqrt{2}\Omega$ but greater than $50\sqrt{2}\Omega$

(C) less than $50\sqrt{2}\Omega$ but greater than $25\sqrt{2}\Omega$

(D) less than $25\sqrt{2}\Omega$ but greater than 0

EM0115

83. A simple pendulum has a time period T if there is no air resistance. If a small air resistance is acting on the bob as it oscillates,

(A) The time period will be initially more than T and decreases with time.

(B) The time period will be less than T initially and increases with time

(C) The time period will be less than T and remains constant

(D) The time period will be more than T and remains constant.

EM0117

84. A pendulum with time period of 1s is losing energy due to damping. At certain time its energy is 45 J. If after completing 15 oscillations, its energy has become 15 J, its damping constant (in s^{-1}) is:- **[JEE Mains On line 2015]**

(A) 2

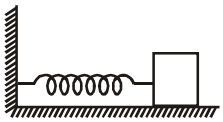
(B) $\frac{1}{15}\ln 3$

(C) $\frac{1}{2}$

(D) $\frac{1}{30}\ln 3$

EM0125

85. In the situation shown, the block can execute free oscillation (no damping) with angular frequency ω_1. In presence of weak damping, it executes damped SHM with angular frequency ω_2. When it is subjected to a sinusoidal force, it executes forced oscillation with maximum amplitude at angular frequency ω_3 (assume damping is present) :-



(A) $\omega_1 > \omega_2 > \omega_3$

(B) $\omega_1 > \omega_2 = \omega_3$

(C) $\omega_1 = \omega_2 = \omega_3$

(D) $\omega_1 > \omega_3 > \omega_2$

EM0123

86. In forced oscillation of a particle the amplitude is maximum for a frequency ω_1 of the force, while the power is maximum for a frequency ω_2 of the force; then – **[AIEEE- 2004]**

(A) $\omega_1 = \omega_2$

(B) $\omega_1 > \omega_2$

(C) $\omega_1 < \omega_2$ when damping is small and $\omega_1 > \omega_2$ when damping is large

(D) $\omega_1 < \omega_2$

EM0124

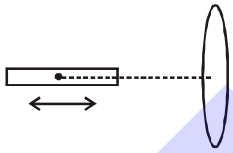
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

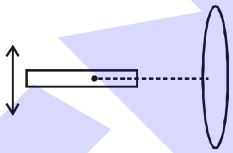
EXERCISE (O-2)

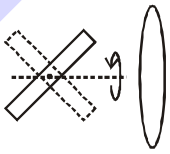
SINGLE CORRECT TYPE QUESTIONS

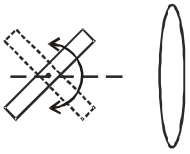
A. *Faraday law and Lenz law*

1. A magnetic bar is in front of a coil as shown below. The line joining the center of the bar and the center of the coil (central axis) is perpendicular to the plane of the coil. Which of the following motions of the bar will **NOT** induce electric current in the coil?

(A) Translational motion back and forth. 

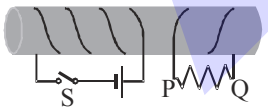
(B) Translational motion up and down. 

(C) The bar axis is at an angle to the central axis and spins around it. 

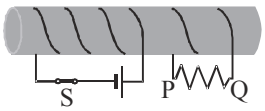
(D) The bar axis swings back and forth about the central axis within the paper plane 

EM0358

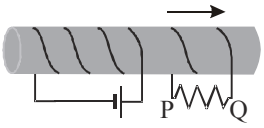
2. For each of the experiments (1, 2, 3, 4) shown in figure. Choose the **CORRECT** option(s) which shows the direction of current flow through the resistor PQ? Note that the wires are not always wrapped around the plastic tube in the same way.



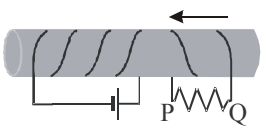
(1) S to be closed



(2) S to be opened



(3) Resistor coil PQ moves to right



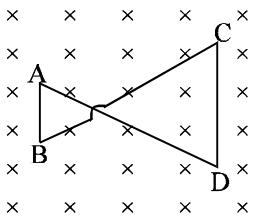
(4) Resistor coil PQ moves to left

(1)	**(2)**	**(3)**	**(4)**
(A) P to Q	P to Q	P to Q	P to Q
(B) P to Q	Q to P	P to Q	Q to P
(C) Q to P	Q to P	Q to P	Q to P
(D) Q to P	Q to P	P to Q	P to Q

EM0141

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

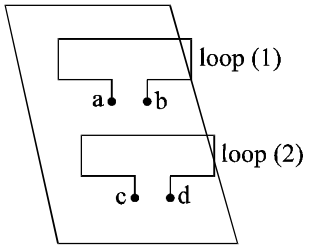
3. A conducting wire frame is placed in a magnetic field which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced currents in wires AB and CD are



(A) B to A and D to C (B) A to B and C to D
(C) A to B and D to C (D) B to A and C to D

EM0059

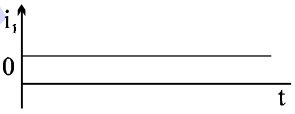
4. An electric current i_1 can flow either direction through loop (1) and induced current i_2 in loop (2). Positive i_1 is when current is from 'a' to 'b' in loop (1) and positive i_2 is when the current is from 'c' to 'd' in loop (2) In an experiment, the graph of i_2 against time 't' is as shown below

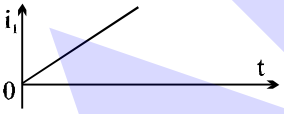


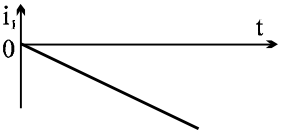


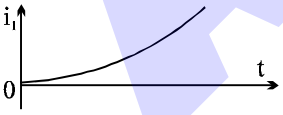
Which one(s) of the following graphs could have caused i_2 to behave as given above.

(A) 

(B) 

(C) 

(D) 

(E) 

EM0159

B. Motional EMF

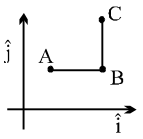
5. A thin wire of length 2m is perpendicular to the xy plane. It is moved with velocity $\vec{v} = (2\hat{i} + 3\hat{j} + \hat{k})\,m/s$ through a region of magnetic induction $\vec{B} = (\hat{i} + 2\hat{j})\ Wb/m^2$. Then potential difference induced between the ends of the wire :

(A) 2 volts (B) 4 volts (C) 0 volts (D) none of these

EM0061

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\3-Magnetic Effect of Current-EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

6. There is a uniform magnetic field B normal to the xy plane. A conductor ABC has length AB = l_1, parallel to the x-axis, and length BC = l_2, parallel to the y-axis. ABC moves in the xy plane with velocity $v_x\hat{i} + v_y\hat{j}$. The potential difference between A and C is proportional to :-



(A) $v_x l_1 + v_y l_2$ (B) $v_x l_2 + v_y l_1$ (C) $v_x l_2 - v_y l_1$ (D) $v_x l_1 - v_y l_2$

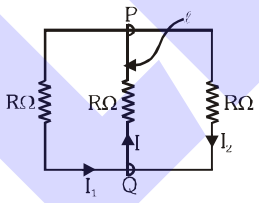
EM0063

7. Two parallel long straight conductors lie on a smooth surface. Two other parallel conductors rest on them at right angles so as to form a square of side a initially. A uniform magnetic field B exists at right angles to the plane containing the conductors. They all start moving out with a constant velocity v. If r is the resistance per unit length of the wire the current in the circuit will be

(A) $\frac{Bv}{r}$ (B) $\frac{Br}{v}$ (C) Bvr (D) Bv

EM0153

8. A rectangular loop has a sliding connector PQ of length ℓ and resistance RΩ and it is moving with a speed v as shown. The set-up is placed in a uniform magnetic field going into the plane of the paper. The three currents I_1, I_2 and I are :- **[AIEEE - 2010]**

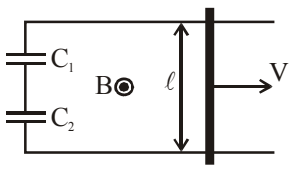


(A) $I_1 = I_2 = \frac{B\ell v}{6R}$, $I = \frac{B\ell v}{3R}$ (B) $I_1 = -I_2 = \frac{B\ell v}{R}$, $I = \frac{2B\ell v}{R}$

(C) $I_1 = I_2 = \frac{B\ell v}{3R}$, $I = \frac{2B\ell v}{3R}$ (D) $I_1 = I_2 = I = \frac{B\ell v}{R}$

EM0176

9. Two parallel conductors located at a distance $\ell = 0.5$ m from each other, is moved with constant speed of v = 10 m/s. At the left end of the conductors, capacitors C_1 and C_2 are connected in series as shown such that $C_2 = 1.5C_1$. The whole system is in a uniform magnetic field perpendicular to the plane of the system. Find the magnetic field, if the voltage across the capacitor C_2 is 0.5 V :-

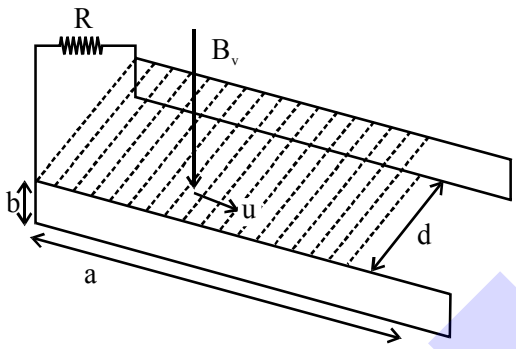


(A) 0.5 T (B) 0.25 T (C) 0.125 T (D) 1 T

EM0359

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects\Current\EM\8.AC\Eng\02_EM\8.AC\02_Exercises.p65

10. The figure shows an apparatus suggested by Faraday to generate electric current from a flowing river. Two identical conducting plates of length a and width b are placed parallel facing one another on opposite sides of the river following with velocity u at a distance d apart. Now both the plates are connected by a load resistance R. Then the current through the load R is :- (Consider vertical component of the magnetic field produced by earth is B_v and the resistivity of river water is ρ.)

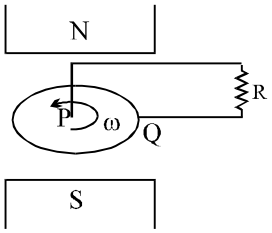


(A) $\frac{B_v ub}{R}$ (B) $\frac{B_v ub}{R+\frac{\rho d}{ab}}$ (C) $\frac{B_v ud}{R+\frac{\rho d}{ab}}$ (D) None of the above

EM0145

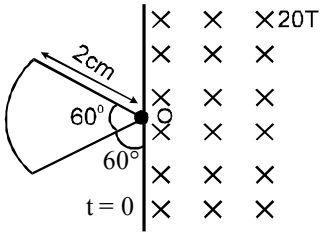
11. A metal disc rotates freely, between the poles of a magnet in the direction indicated. Brushes P and Q make contact with the edge of the disc and the metal axle. What current, if any, flows through R?

(A) a current from P to Q
(B) a current from Q to P
(C) no current, because the emf in the disc is opposed by the back emf
(D) no current, because the emf induced in one side of the disc is opposed by the emf induced in the other side.
(E) no current, because no radial emf is induced in the disc



EM0065

12. A uniform magnetic field 20 T exists on right side of the boundary in a gravity free space as shown in figure. The given circular arc of radius 2 cm made of conducting wire of total resistance 4Ω is rotated around point O at a constant angular speed 2 rad per second. Power required to maintain the constant angular velocity between time interval $t = \frac{\pi}{6}$ s to $t = \frac{\pi}{3}$ s is :-

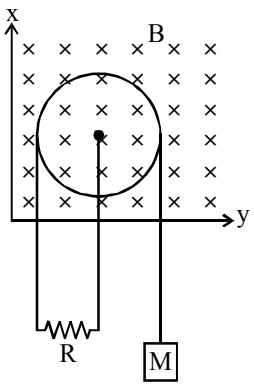


(A) 64 μ W (B) 32 μ W (C) 128 μ W (D) 16 μ W

EM0149

node06\B0BC-BD\Kota\JEE(Advanced)\Enfinity\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

13. The block of mass (M) is connected by thread which is wound on a pulley, free to rotate about fixed horizontal axis as shown. A uniform magnetic field B exists in a horizontal plane. The disc is connected with the resistance R as shown. Calculate the terminal velocity of the block if it was released from rest. Treat pulley as uniform metallic disc of radius L.

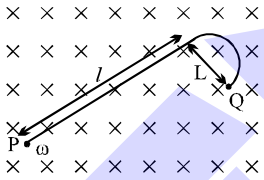


(A) $\frac{4mgR}{B^2L^2}$ (B) $\frac{3mgR}{4B^2L^2}$

(C) $\frac{2mgR}{B^2L^2}$ (D) $\frac{3mgR}{2B^2L^2}$

EM0150

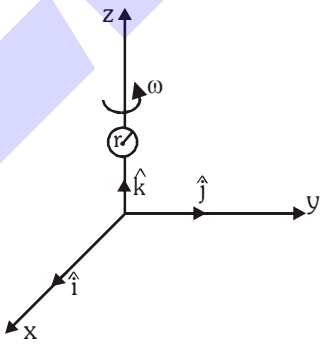
14. When a 'J' shaped conducting rod is rotating in its own plane with constant angular velocity ω, about one of its end P, in a uniform magnetic field $\vec{B}$ directed normally into the plane of paper then magnitude of emf induced across it will be



(A) $B\omega\sqrt{L^2+l^2}$ (B) $\frac{1}{2}B\omega L^2$ (C) $\frac{1}{2}B\omega(L^2+l^2)$ (D) $\frac{1}{2}B\omega l^2$

EM0156

15. A circular loop wire of radius r rotates about the z–axis with angular velocity ω. The normal to the loop is always perpendicular to the z–axis. At time t = 0, the normal is parallel to the y–axis. An external magnetic field $\vec{B} = B_y\hat{j} + B_z\hat{k}$ is applied. The EMF ε(t) induced in the loop is



(A) $\pi r^2\omega B_y \sin\omega t$ (B) $\pi r^2\omega B_z \cos\omega t$ (C) $\pi r^2\omega B_z \sin\omega t$ (D) $\pi r^2\omega B_y \cos\omega t$

EM0148

16. A composite rod of length ℓ is one fourth insulator and remaining conductor is made to rotate freely with angular velocity ω, in a space free of any gravitational, electric & magnetic field. Then potential difference across the conducting region will be (rotation is about insulating end).

(A) $\frac{3m_e\omega^2\ell^2}{4e}$ (B) $\frac{1}{4}\frac{m_e\omega^2\ell^2}{e}$ (C) $\frac{1}{16}\frac{m_e\omega^2\ell^2}{e}$ (D) $\frac{15}{32}\frac{m_e\omega^2\ell^2}{e}$

EM0147

node06\B0BC-BD\Vicky\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects of Current-EM1 & AC\Eng\02_EM1 & AC\02_Exercises.p65

17. A metallic rod of length L and mass M is moving under the action of two unequal forces F_1 and F_2 (directed opposite to each other) acting at its ends along its length. Ignore gravity and any external magnetic field. If specific charge of electrons is (e/m), then the potential difference between the ends of the rod in steady state must be

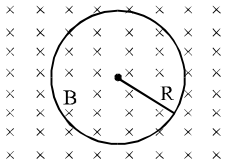
(A) $|F_1 - F_2|mL/eM$ (B) $(F_1 + F_2)\ mL/eM$

(C) $[mL/eM]\ ln\ [F_1/F_2]$ (D) None

EM0152

C. Induced electric field

18. A conducting loop of radius R is present in a uniform magnetic field B perpendicular to the plane of the ring. If radius R varies as a function of time 't', as $R = R_0 + t$. The e.m.f induced in the loop is



(A) $2\pi(R_0 + t)B$ clockwise (B) $\pi(R_0 + t)B$ clockwise

(C) $2\pi(R_0 + t)B$ anticlockwise (D) zero

EM0060

19. **Statement–1 :** For a charged particle moving from point P to point Q the net work done by an induced electric field on the particle is independent of the path connecting point P to point Q.

Statement–2 : The net work done by a conservative force on an object moving along closed loop is zero.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

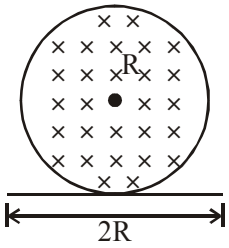
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

EM0071

20. A uniform but time varying magnetic field is present in a circular region of radius R. The magnetic field is perpendicular and into the plane of the loop and the magnitude of field is increasing at a constant rate α. There is a straight conducting rod of length 2R placed as shown in figure. The magnitude of induced emf across the rod is

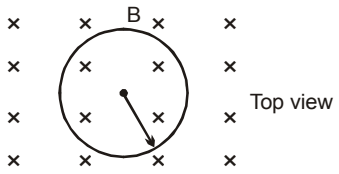


(A) $\pi R^2\alpha$ (B) $\frac{\pi R^2\alpha}{2}$ (C) $\frac{R^2\alpha}{\sqrt{2}}$ (D) $\frac{\pi R^2\alpha}{4}$

EM0069

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\3-Magnetic Effects of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

21. A non conducting ring (of mass m, radius r, having charge Q) is placed on a rough horizontal surface (in a region with transverse magnetic field). The field is increasing with time at the rate R and coefficient of friction between the surface and the ring is μ. For ring to remain in equilibrium μ should be greater than :-



(A) $\frac{QrR}{mg}$ (B) $\frac{QrR}{2mg}$ (C) $\frac{QrR}{3mg}$ (D) $\frac{2QrR}{mg}$

EM0143

22. In figure (a) a solenoid produces a magnetic field whose strength increases into the plane of the page. An induced emf is established in a conduction loop surrounding the solenoid, and this emf lights bulbs A and B. In figure (b) point P and Q are shorted. After the short is inserted

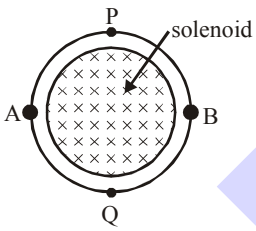


fig. (a)

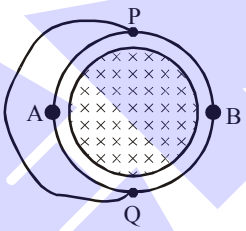


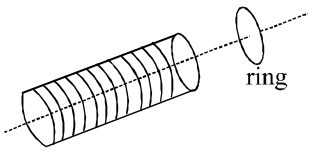
fig. (b)

(A) Bulb A goes out, bulb B gets brighter (B) Bulb B goes out, bulb A gets brighter
(C) Bulb A goes out, bulb B gets dimmer (D) Bulb B goes out, bulb A gets dimmer

EM0158

D. Self inductance and mutual inductance

23. The diagram shows a solenoid carrying time varying current $I = I_0 t$. (I_0 is constant) On the axis of this solenoid a conducting ring is being placed as shown in figure. The mutual inductance of the ring and the solenoid is M and self inductance of the ring is L. If the resistance of the ring is R then the maximum current which can flow through the ring is



(A) $(2M + L)\frac{I_0}{R}$ (B) $\frac{MI_0}{R}$ (C) $(2M - L)\frac{I_0}{R}$ (D) $(M + L)\frac{I_0}{R}$

EM0360

24. The mutual inductance between a primary and secondary circuits is 0.5 H. The resistances of the primary and the secondary circuits are 20 ohms and 5 ohms respectively. To generate a current of 0.4A in the secondary, current in the primary must be changed at the rate of

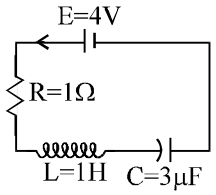
(A) 4.0 *A/s* (B) 16.0 *A/s* (C) 1.6 *A/s* (D) 8.0 *A/s*

EM0361

node06\B0BC-BD\Vicky\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects\Current EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

E. R-L circuit

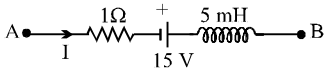
25. The current in the given circuit is increasing with a rate a = 4 amp/s. The charge on the capacitor at an instant when the current in the circuit is 2 amp will be :



(A) 4μC (B) 5μC (C) 6μC (D) none of these

EM0075

26. The network shown in the figure is part of a complete circuit. If at a certain instant, the current I is 5A and it is decreasing at a rate of 10^3 As^{-1} then $V_B - V_A$ equals



(A) 20 V (B) 15 V (C) 10 V (D) 5 V

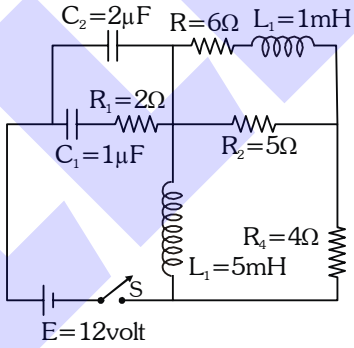
EM0077

27. In **Problem 26**, if I is reversed in direction, then $V_B - V_A$ equals

(A) 5 V (B) 10 V (C) 15 V (D) 20 V

EM0078

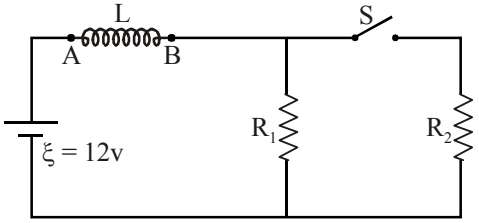
28. Find current through the battery just after switch S is closed. Initially all the capacitors are uncharged:



(A) $\frac{4}{3}$A (B) $\frac{5}{3}$A (C) 0 A (D) $\frac{3}{5}$A

EM0362

29. The circuit shown has been operating for a long time. The instant after the switch in the circuit labeled S is opened, what is the voltage across the inductor V_L and which labeled point (A or B) of the inductor is at a higher potential ? Take R_1 = 4.0 Ω, R_2 = 8.0 Ω, and L = 2.5 H.

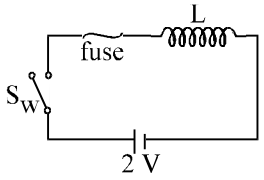


(A) V_L = 12 V ; Point A is at the higher potential
(B) V_L = 12 V ; Point B is at the higher potential
(C) V_L = 6V ; Point A is at the higher potential
(D) V_L = 6V ; Point B is at the higher potential

EM0161

node06\B0BC-BD\Kota\JEE(Advanced)\Enfinity\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

30. In the circuit shown, the cell is ideal. The coil has an inductance of 4H and zero resistance. F is a fuse of zero resistance and will blow when the current through it reaches 5A. The switch is closed at t = 0. The fuse will blow :



(A) just after t = 0 (B) after 2s (C) after 5s (D) after 10s

EM0160

31. For L-R circuit, the time constant is equal to

(A) twice the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

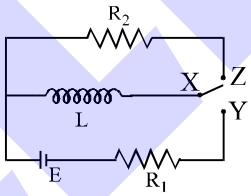
(B) ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

(C) half the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

(D) square of the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

EM0088

32. In the circuit shown, X is joined to Y for a long time, and then X is joined to Z. The total heat produced in R_2 is :



(A) $\frac{LE^2}{2R_1^2}$ (B) $\frac{LE^2}{2R_2^2}$ (C) $\frac{LE^2}{2R_1R_2}$ (D) $\frac{LE^2R_2}{2R_1^2}$

EM0091

F. L-C oscillation

33. A fully charged capacitor C with intial charge q_0 is connected to a coil of self inductance L at t = 0. The time at which the energy is stored equally between the electric and the magnetic fields is :-

[AIEEE - 2011]

(A) $2\pi\sqrt{LC}$ (B) $\sqrt{LC}$ (C) $\pi\sqrt{LC}$ (D) $\frac{\pi}{4}\sqrt{LC}$

EM0180

G. AC circuits

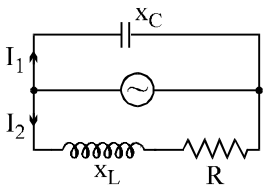
34. The effective value of current $i = 2\sin 100\pi t + 2\sin(100\pi t + 30°)$ is :

(A) $\sqrt{2}$ Amp (B) $2\sqrt{2+\sqrt{3}}$ Amp (C) 4 Amp (D) None

EM0103

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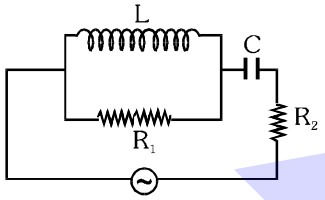
35. In the shown AC circuit phase difference between currents I_1 and I_2 is



(A) $\frac{\pi}{2} - \tan^{-1}\frac{x_L}{R}$ (B) $\tan^{-1}\frac{x_L - x_C}{R}$ (C) $\frac{\pi}{2} + \tan^{-1}\frac{x_L}{R}$ (D) $\tan^{-1}\frac{x_L - x_C}{R} + \frac{\pi}{2}$

EM0107

36. For the given circuit



(A) The phase difference between I_L & I_{R_1} is 0°

(B) The phase difference between V_C & V_{R_2} is 90°

(C) The phase difference between I_L & I_{R_1} is 180°

(D) The phase difference between V_C & V_{R_2} is 180°

EM0363

37. An AC voltage source of variable angular frequency ω and fixed amplitude V_0 is connected in series with a capacitance C and an electric bulb of resistance R (inductance zero). When ω is increased

[JEE 2010]

(A) the bulb glows dimmer (B) the bulb glows brighter

(C) total impedance of the circuit is unchanged (D) total impedance of the circuit increases

EM0194

38. A series RLC circuit is driven by a generator at frequency 1000 Hz. The inductance is 90.0 mH; capacitance is 0.500 μF; and the phase constant has a magnitude of 60.0° (Take: $\pi^2 = 10$) Choose **INCORRECT** statement :-

(A) Here current leads the voltage in phase

(B) Here voltage leads the current in phase

(C) Resistance of circuit is $\frac{80\pi}{\sqrt{3}}\Omega$

(D) At resonance $\omega = \frac{\sqrt{2}}{3}\times 10^4$ rad/sec.

EM0364

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\3-Magnetic Effect of Current-EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

39. In a series LCR circuit R = 200Ω and the voltage and the frequency of the main supply is 220 V and 50 Hz respectively. On taking out the capacitance from the circuit the current lages behind the voltage by 30°. On taking out the inductor from the circuit the current leads the voltage by 30°. The power dissipated in the LCR circuit is : **[AIEEE - 2010]**

(A) 242 W (B) 305 W (C) 210 W (D) 0 W

EM0177

40. The primary of a 1 : 3 step - up transformer is connected to a source and the secondary is connected to a resistor R. The power dissipated by R in this situation is P. If R is connected directly to the source it will dissipate a power of :

(A) P/9 (B) P/3 (C) P (D) 3P

EM0109

H. Damped and Forced oscillation

41. Five particles undergodamped harmonic motion. Values for the spring constant k, the damping constant b, and the mass m are given below. Which leads to the smallest rate of loss of mechanical energy at the initial moment?

(A) k = 100N/m, m = 50g, b = 8g/s (B) k = 150N/m, m = 50g, b = 5g/s

(C) k = 150N/m, m = 10g, b = 8g/s (D) k = 200N/m, m = 8g, b = 6g/s

EM0113

42. A block is executing damped harmonic oscillation with time period T. Choose correct statement

(1) Time taken to go from extreme to mean position is $\frac{T}{4}$

(2) Time taken to go from one extreme to another is $\frac{T}{2}$

(3) Time taken to go from one extreme to another is less than $\frac{T}{2}$

(4) Time taken to go from one extreme to another is more than $\frac{T}{2}$

(A) 1,2 only (B) 1,2,3 only (C) 2 only (D) 1,2,4 only

EM0118

43. The angular frequecny of the damped oscillator is given by, $\omega = \sqrt{\frac{k}{m} - \frac{r^2}{4m^2}}$ where k is the spring constant, m is the mass of the oscillator and r is damping constant. If the ratio $\frac{r^2}{mk}$ is 8%, the change in time period composed to the undamped oscillator is approximately as follows:

(A) decreases by 8% (B) decreases by 1% (C) increases by 1% (D) increases by 8%

EM0119

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\3-Magnetic Effect of Current-EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

44. Two spheres of the same diameter but of different masses are suspended by strings of equal length. If the spheres are deflected from their positions of equilibrium, which of the two will have a greater oscillation period and which will have a greater logarithmic decrement if their oscillations occur in a real medium with viscosity?

(A) Heavier mass has larger time period & greater logrithmic decrement

(B) Lighter mass has larger time period & greater logrithmic decrement

(C) Lighter mass has larger time period but lesser logrithmic decrement

(D) Heavier mass has larger time period but lesser logrithmic decrement

EM0120

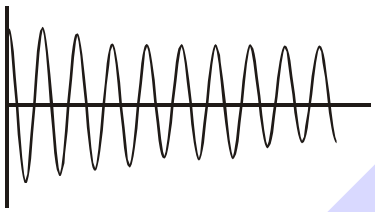
45. The amplitude of a simple pendulum, oscillating in air with a small spherical bob, decreases from 10cm to 8 cm in 40 seconds. Assuming that Stokes law is valid, and ratio of the coefficient of viscosity of air to that of carbon dioxide is 1.3, the time in which amplitude of this pendulum will reduce from 10 cm to 5 cm in carbondioxide will be close to (ℓn 5 = 1.601, ℓn 2 = 0.693).

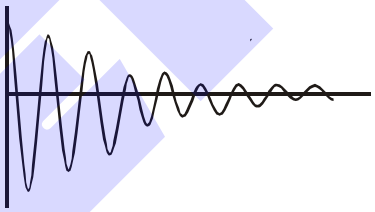
[JEE Main Online 2014]

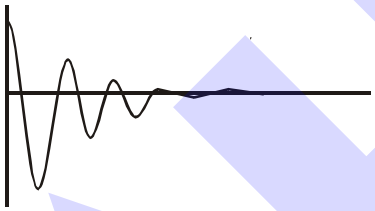
(A) 231 s (B) 208 s (C) 142 s (D) 161 s

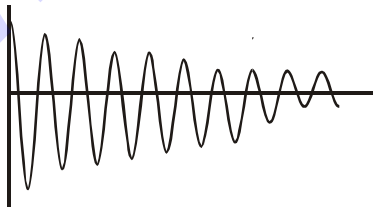
EM0121

46. Which graph has the highest Q factor ?

(A) 

(B) 

(C) 

(D) 

EM0122

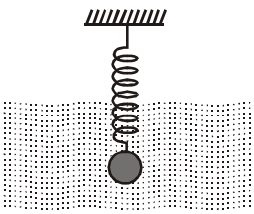
47. A ball is oscillating in a viscous medium. If we use a ball of same radius but higher density.

(A) Time period is more but damping coefficient $\left(\frac{b}{2m}\right)$ is less.

(B) Time period is less but damping coefficient $\left(\frac{b}{2m}\right)$ is more

(C) Time period as well as damping coefficient $\left(\frac{b}{2m}\right)$ is more

(D) Time period as well as damping coefficient $\left(\frac{b}{2m}\right)$ is less.



EM0365

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

48. A coil is suspended in a uniform magnetic field, with the plane of the coil parallel to the magnetic lines of force. When a current is passed through the coil it starts oscillating; it is very difficult to stop. But if an aluminium plate is placed near to the coil, it stops. This is due to :- **[AIEEE - 2012]**

(A) Electromagnetic induction in the aluminium plate giving rise to electromagnetic damping

(B) Development of air current when the plate is placed

(C) Induction of electrical charge on the plate

(D) Shielding of magnetic lines of force as aluminium is a paramagnetic material

EM0181

49. A body is connected to a spring and subjected to fluid friction so that its motion is overdamped. If the body is pulled away from the equilibrium position and projected with an initial velocity towards the equilibrium,

(A) It can never cross the equilibrium position

(B) It can cross the equilibrium position only once that too only if the initial velocity is sufficiently high

(C) If the initial velocity is sufficiently high, it can oscillate about the equilibrium position a few times and come to rest at the equilibrium position

(D) If the initial velocity is sufficiently high, it can oscillate about the equilibrium position a maximum of two times and come to rest at the equilibrium position

EM0366

50. If a simple pendulum has Significant amplitude (up to a factor of 1/ e of original) only in the period between t = 0s to t = τs, then τ may be called the average life ofthe pendulum. When the spherical bob of the pendulum suffers a retardation (due to viscous drag) proportional to its velocity, with 'b' as the constant of proportionality, the average lifetime ofthe pendulum is (assuming damping is small) in seconds:

[AIEEE-2012]

(A) $\frac{1}{b}$ (B) $\frac{2}{b}$ (C) $\frac{0.693}{b}$ (D) b

EM0182

51. Two underdamped oscillators are known to have the same natural frequency ω_0. The mass and damping coefficient of the first oscillator are m_1 and b_1, and the mass and damping coeficient of the second oscillator are m_2 and b_2, respectively. A sinusoidal driving force of $F_{ext} = F_0 \cos \omega t$ is applied to each oscillator. Starting with ω far from ω_0 the driving force is tuned in order to observe resonant behavior. If $m_1 = 4\, m_2$ and $b_1 = 2b_2$, then which one of the following statements concerning the driven oscillations is correct ?

(A) The resonant peak of the first driven oscillator is higher and narrower than that ofthe second oscillator.

(B) The resonant peak of the first driven oscillator is higher and wider than that ofthe second oscillator.

(C) The resonant peak of the first driven oscillator is lower and wider than that ofthe second oscillator.

(D) The resonant peak of the first driven oscillator is lower and narrower than that ofthe second oscillator.

EM0116

node06\B0BC-BD\Work\JEE\Advanced\Enthuse\Phy\Module\3-Magnetic Effects of Current\EM\8.AC\Eng\02_EM8.AC\02_Exercises.p65

52. Oscillator is moving according to the law, $x = x_0 \sin \omega t$ because of a driving force $F = F_0 \cos \omega t$ acting on it. What is the damping coefficient γ of the oscillator ? Mass of the oscillator is m. The resistive force is dependent on velocity v as $F = -\gamma mv$.

(A) $\frac{F_o}{x_o \omega}$ (B) $\frac{F_o}{2x_o \omega}$ (C) $\frac{F_o}{mx_o \omega}$ (D) $\frac{F_o}{2mx_o \omega}$

EM0367

53. Figure (a) shows plot of voltage across the capacitor as a function of the driving frequency for a sinusoidally driven electromagnetic oscillator LCR circuit. Figure (b) shows phase angle ϕ (phase difference between voltage and current) vs ω/ω_0 graph for same circuit, for three different quality factor graph 1, 2, 3 of figure (a) and each one can be matched by one of graphs a, b, c also of figure (b) :-

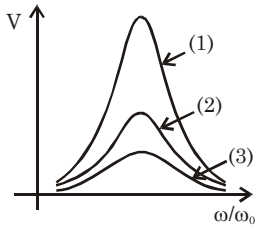


figure (a)

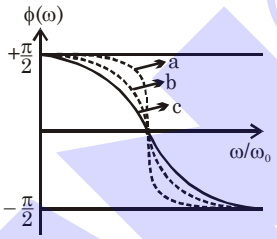


figure (b)

(A) Graph (3) corresponds to graph (a)
(B) Graph (1) corresponds to graph (c)
(C) The circuit of graph 1 has high quality factor.
(D) The circuit of graph 3 has high quality factor.

EM0368

MULTIPLE CORRECT TYPE QUESTIONS

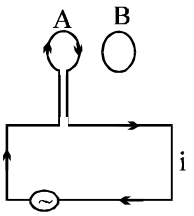
A. *Faraday law and Lenz law*

54. The dimension of the ratio of magnetic flux and the resistance is equal to that of :

(A) induced emf (B) charge (C) inductance (D) current

EM0126

55. Two circular coils A and B are facing each other as shown in figure. The current i through A can be altered



(A) there will be repulsion between A and B if i is increased
(B) there will be attraction between A and B if i is increased
(C) there will be neither attraction nor repulsion when i is changed
(D) attraction or repulsion between A and B depends on the direction of current. It does not depend whether the current is increased or decreased.

EM0127

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

56. A bar magnet is moved along the axis of copper ring placed far away from the magnet. Looking from the side of the magnet, an anticlockwise current is found to be induced in the ring. Which of the following may be true?

(A) The south pole faces the ring and the magnet moves towards it.

(B) The north pole faces the ring and the magnet moves towards it.

(C) The south pole faces the ring and the magnet moves away from it.

(D) The north pole faces the ring and the magnet moves away from it.

EM0128

57. A current carrying infinitely long wire is kept along the diameter of a circular wire loop, without touching it. The correct statement(s) is (are) **[JEE 2012]**

(A) The emf induced in the loop is zero if the current is constant.

(B) The emf induced in the loop is infinite if the current is constant.

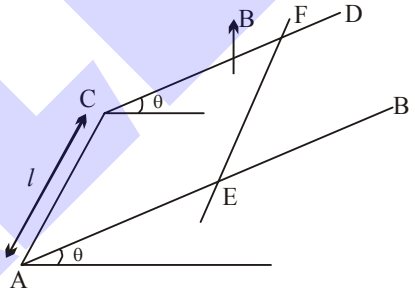
(C) The emf induced in the loop is zero if the current decreases at a steady rate.

(D) The emf induced in the loop is finite if the current decreases at a steady rate.

EM0202

B. Motional EMF

58. AB and CD are smooth parallel rails, separated by a distance l, and inclined to the horizontal at an angle θ. A uniform magnetic field of magnitude B, directed vertically upwards, exists in the region. EF is a conductor of mass m, carrying a current i. For EF to be in equilibrium,



(A) i must flow from E to F

(B) Bil = mg tan θ

(C) Bil = mg sin θ

(D) Bil = mg

EM0129

59. In the **previous question**, if B is normal to the plane of the rails

(A) Bil = mg tan θ

(B) Bil = mg sin θ

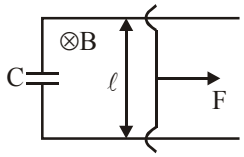
(C) Bil = mg cos θ

(D) equilibrium cannot be reached

EM0130

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EM1 & AC\Eng\02_EM1 & AC\02_Exercises.p65

60. A conducting wire of length ℓ and mass m can slide without friction on two parallel rails and is connected to capacitance C initial charge zero. Whole system lies in a magnetic field B and a constant force F is applied to the rod at t = 0. If initial speed of rod is zero. Then



(A) Speed of the rod at time $t = \dfrac{Ft}{B^2\ell^2C + m}$

(B) The rod moves with constant acceleration

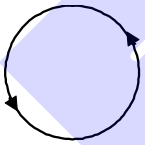
(C) The charge of capacitor varies linearly with velocity

(D) Charge on the capacitor will never attain steady state value.

EM0369

C. Induced electric field

61. In the given diagram, a line of force of a particular force field is shown. Out of the following options, it can never represent **[JEE 2006]**



(A) an electrostatic field

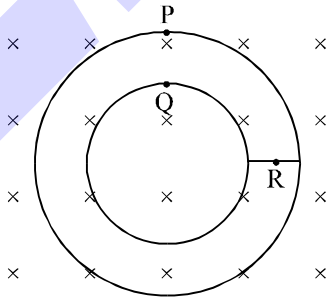
(B) a magnetostatic field

(C) a gravitational field of a mass at rest

(D) an induced electric field

EM0370

62. Figure shown plane figure made of a conductor located in a magnetic field along the inward normal to the plane of the figure. The magnetic field starts diminishing. Then the induced current



(A) at point P is clockwise

(B) at point Q is anticlockwise

(C) at point Q is clockwise

(D) at point R is zero

EM0168

E. R-L circuit

63. An LR circuit with a battery is connected at t = 0. Which of the following quantities is not zero just after the circuit

(A) current in the circuit

(B) magnetic field energy in the inductor

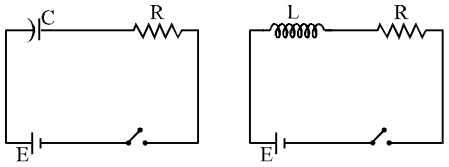
(C) power delivered by the battery

(D) emf induced in the inductor

EM0132

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

64. The switches in figures (a) and (b) are closed at t = 0

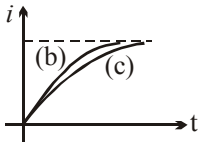


(a) (b)

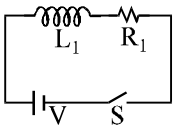
(A) The charge on C just after t = 0 is EC.
(B) The charge on C long after t = 0 is EC.
(C) The current in L just after t = 0 is E/R.
(D) The current in L long after t = 0 is E/R.

EM0133

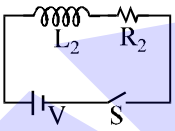
65. Current growth in two L-R circuits (b) and (c) as shown in figure (a). Let L_1, L_2, R_1 and R_2 be the corresponding values in two circuits. Then :-



(a)



(b)

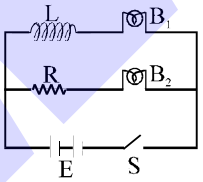


(c)

(A) $R_1 > R_2$ (B) $R_1 = R_2$ (C) $L_1 > L_2$ (D) $L_1 < L_2$

EM0134

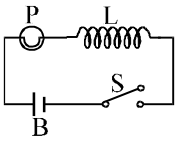
66. An inductor L, a resistance R and two identical bulbs B_1 and B_2 are connected to a battery through a switch S as shown in the figure. The resistance of coil having inductance L is also R. Which of the following statement gives the correct description of the happenings when the switch S is closed?



(A) The bulb B_2 lights up earlier than B_1 and finally both the bulbs shine equally bright.
(B) B_1 lights up earlier and finally both the bulbs acquire equal brightness.
(C) B_2 lights up earlier and finally B_1 shines brighter than B_2.
(D) B_1 and B_2 light up together with equal brightness all the time.

EM0135

67. In figure, a lamp P is in series with an iron-core inductor L. When the switch S is closed, the brightness of the lamp rises relatively slowly to its full brightness than it would do without the inductor. This is due to



(A) the low resistance of P (B) the induced-emf in L
(C) the low resistance of L (D) the high voltage of the battery B

EM0136

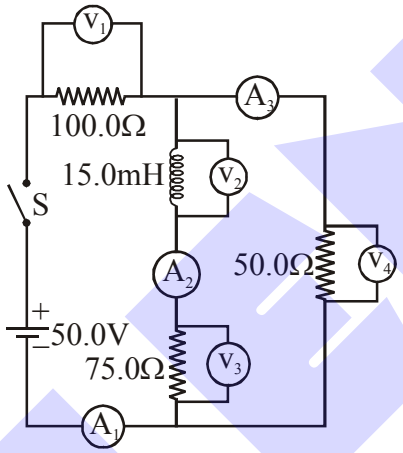
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects of Current\EM\E.AC\Eng\02_EM\E.AC\02_Exercises.p65

68. Two different coils have self inductance 8mH and 2mH. The current in one coil is increased at a constant rate. The current in the second coil is also increased at the same constant rate. At a certain instant of time, the power given to the two coils is the same. At that time the current, the induced voltage and the energy stored in the first coil are I_1, V_1 and W_1 respectively. Corresponding values for the second coil at the same instant are I_2, V_2 and W_2 respectively. Then:

(A) $\frac{I_1}{I_2} = \frac{1}{4}$ (B) $\frac{I_1}{I_2} = 4$ (C) $\frac{W_2}{W_1} = 4$ (D) $\frac{V_2}{V_1} = \frac{1}{4}$

EM0137

69. In the circuit shown in figure, just after switch S is closed, the reading in each ammeter (in amp) and voltmeter (in V) will be :-



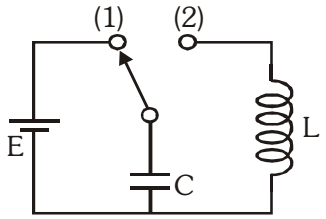
(A) $A_1 = \frac{1}{3}, A_2 = 0$ & $A_3 = \frac{1}{3}$ (B) $A_1 = \frac{5}{13}, A_2 = \frac{2}{13}$ & $A_3 = \frac{3}{13}$

(C) $V_1 = \frac{100}{3}, V_3 = 0$ (D) $V_2 = \frac{50}{3}, V_4 = \frac{100}{3}$

EM0371

F. L-C oscillation

70. Initially key was placed on (1) till the capacitor got fully charged. Key is placed on (2) at t = 0. The time when the energy in both capacitor and inductor will be same-

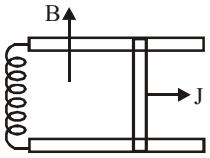


(A) $\frac{\pi\sqrt{LC}}{4}$ (B) $\frac{\pi\sqrt{LC}}{2}$ (C) $\frac{5\pi\sqrt{LC}}{4}$ (D) $\frac{5\pi\sqrt{LC}}{2}$

EM0138

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

71. Two parallel resistanceless rails are connected by an inductor of inductance L at one end as shown in the figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length ℓ and mass m is placed transverse on the rails and given an impulse J towards the rightward direction. Then choose the **CORRECT** option (s).



(A) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{3J^2L}{4B^2\ell^2m}}$

(B) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{4Lm}}$

(C) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{3J^2L}{B^2\ell^2m}}$

(D) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{mL}}$

EM0167

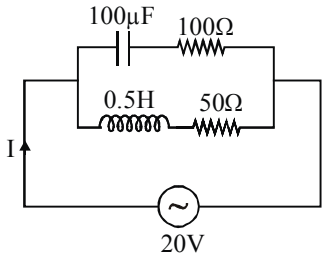
G. AC circuits

72. A series R-C circuit is connected to AC voltage source. Consider two cases ; (A) when C is without a dielectric medium and (B) when C is filled with dielectric of constant 4. The current I_R through the resistor and voltage V_C across the capacitor are compared in the two cases. Which of the following is/are true? **[JEE 2011]**

(A) $I_R^A > I_R^B$ (B) $I_R^A < I_R^B$ (C) $V_C^A > V_C^B$ (D) $V_C^A < V_C^B$

EM0198

73. In the given circuit, the AC source has $\omega = 100$ rad/s. Considering the inductor and capacitor to be ideal, the correct choice (s) is(are) **[JEE 2012]**

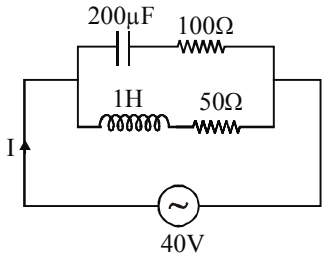


(A) The current through the circuit, I is 0.3 A. (B) The current through the circuit, i is 0.3 $\sqrt{2}$A.
(C) The voltage across 100Ω resistor = $10\sqrt{2}$V. (D) The voltage across 50Ω resistor = 10V.

EM0203

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects of Current-EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

74. In the given circuit, the AC source has $\omega = 50$ rad/s. Considering the inductor and capacitor to be ideal, the **CORRECT** choice (s) is(are) :-



(A) The voltage across 100Ω resistor = $20\sqrt{2}$V.
(B) The voltage across 50Ω resistor = $20\sqrt{2}$V.
(C) The current through the circuit, I is $\frac{2}{\sqrt{10}}$A.
(D) The current through the circuit, I is 1.2A.

EM0372

H. Damped and Forced oscillation

75. A Mass-Spring-Damper system is tested to determine the value of the viscous damping coefficient b. Assume k = 161N/m and m = 4 kg. The vibrational amplitude is observed to decrease to 1/e of its initial value after two consecutive cycles.
(A) its logarithmic decrement is 0.5
(B) angular frequency of its oscillation is $\sqrt{\frac{161}{4}}$rad / s
(C) time period of its oscillation is 1 sec
(D) b = 4 kg/sec

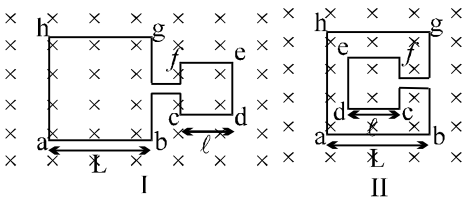
EM0373

COMPREHENSION TYPE QUESTIONS

A. Faraday law and Lenz law

Paragraph for Question No. 76 to 79

The adjoining figure shows two different arrangements in which two square wire frames of same resistance are placed in a uniform constantly decreasing magnetic field B.



76. The value of magnetic flux in each case is given by
(A) Case I: $\Phi = \pi(L^2 + \ell^2)B$; Case II: $\Phi = \pi(L^2 - \ell^2)B$
(B) Case I: $\Phi = \pi(L^2 + \ell^2)B$; Case II: $\Phi = \pi(L^2 + \ell^2)B$
(C) Case I: $\Phi = (L^2 + \ell^2)B$; Case II: $\Phi = (L^2 - \ell^2)B$
(D) Case I: $\Phi = (L + \ell)^2B$; Case II: $\Phi = \pi(L - \ell)^2B$

EM0140

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

77. The direction of induced current in the case I is
(A) from a to b and from c to d
(B) from a to b and from *f* to e
(C) from b to a and from d to c
(D) from b to a and from e to *f*

EM0140

78. The direction of induced current in the case II is
(A) from a to b and from c to d
(B) from b to a and from *f* to e
(C) from b to a and from c to d
(D) from a to b and from d to c

EM0140

79. If I_1 and I_2 are the magnitudes of induced current in the cases I and II, respectively, then
(A) $I_1 = I_2$
(B) $I_1 > I_2$
(C) $I_1 < I_2$
(D) nothing can be said

EM0140

C. Induced electric field

Paragraph for Question Nos. 80 to 82

The fact that a changing magnetic flux produces an electric field is basic to the operation of many high energy particle accelerators. Since the principle was first successfully applied to the acceleration of electrons (or β particles) in a device called the betatron, this method of acceleration is often given that name. The general idea involved is shown in figure.

An electromagnet is used to produce a changing flux through a circular loop defined by the doughnut-shaped vacuum chamber. We see that there will be an electric field E along the circular length of the doughnut, i.e. circling the magnet poles, given by

$$2\pi aE = \frac{d\phi}{dt}$$

Where a is the radius of the doughnut. Any charged particle inside the vacuum chamber will experience a force qE and will accelerate. Ordinarily, the charged particle would shoot out of the vacuum chamber and becomes lost.

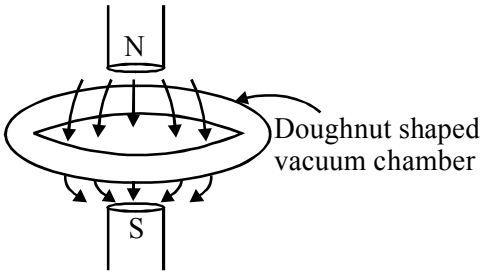
However, if the magnetic field at the position of the doughnut is just proper to satisfy the relation Centripetal force = magnetic force

$$\text{or } \frac{mv^2}{a} = qvB$$

then the charge will travel in a circle within the doughnut. By proper shaping of the magnet pole pieces, this relation can be satisfied. As a result, the charge will move at high speed along the loop within the doughnut. Each time it goes around the loop, it has, in effect, fallen through a potential difference equal to the induced, emf, namely $\varepsilon = \frac{d\phi}{dt}$. Its energy after n trips around the loop will be

q (nε)

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects of Current-EM1 & AC\Eng\02_EM1 & AC\02_Exercises.p65



80. Working of betatron is not based upon which of the following theories :-
(A) Changing magnetic flux induces electric field
(B) Charged particle at rest can be accelerated only by electric fields
(C) magnetic fields can apply a force on moving charges which is perpendicular to both magnetic field and motion of the particle
(D) Beta particles are emitted in radioactive decay process.

EM0139

81. Variable magnetic flux :-
(A) Can change sinusoidally
(B) Should increase all the time
(C) Must becomes zero when induced field is maximum
(D) None of these

EM0139

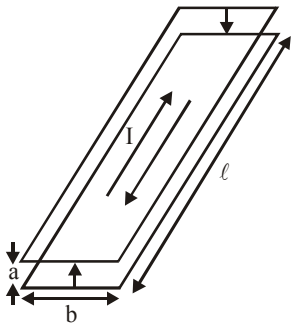
82. Magnetic field which keeps the particles in circular path must :-
(A) Remain a constant every where
(B) Increase gradually which is proportional to K.E. of the particle
(C) Increase gradually which is proportional to speed of the particle
(D) None of these

EM0139

D. Self inductance and mutual inductance

Paragraph for Questions Nos. 83 & 84

Two parallel rectangular superconducting plates of length ℓ, width b and separation a ($\ell >> b >> a$) are joined at each end to form a one-turn coil of negligible resistance. The steady state current is I.



83. The energy stored in magnetic field will be :-

(A) $U = \frac{\mu_0 \ell a I^2}{b}$ (B) $U = \frac{\mu_0 a b I^2}{2\ell}$ (C) $U = \frac{\mu_0 \ell b I^2}{2a}$ (D) $U = \frac{\mu_0 \ell a I^2}{2b}$

EM0374

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

84. The magnitude of the force per unit area between the two plates :-

(A) $\frac{\mu_0 I^2 \ell}{2b}$ (B) $\frac{\mu_0 I^2}{2a^2}$

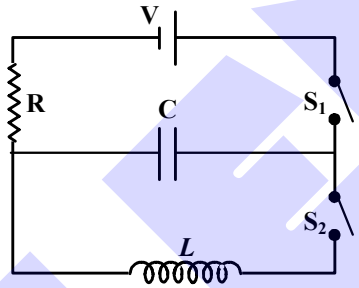
(C) $\frac{\mu_0 I^2}{2b^2}$ (D) $\frac{\mu_0 I^2}{ab}$

EM0374

F. L-C oscillation

Paragraph for Questions no. 85 to 87

The capacitor of capacitance C can be charged (with the help of a resistance R) by a voltage source V, by closing switch S_1 while keeping switch S_2 open. The capacitor can be connected in series with an inductor 'L' by closing switch S_2 and opening S_1.



85. Initially, the capacitor was uncharged. Now, switch S_1 is closed and S_2 is kept open. If time constant of this circuit is τ, then **[JEE 2006]**

(A) after time interval τ, charge on the capacitor is CV/2

(B) after time interval 2τ, charge on the capacitor is $CV(1-e^{-2})$

(C) the work done by the voltage source will be half of the heat dissipated when the capacitor is fully charged.

(D) after time interval 2τ, charge on the capacitor is $CV(1-e^{-1})$

EM0375

86. After the capacitor gets fully charged, S_1 is opened and S_2 is closed so that the inductor is connected in series with the capacitor. Then, **[JEE 2006]**

(A) at t = 0, energy stored in the circuit is purely in the form of magnetic energy

(B) at any time t > 0, current in the circuit is in the same direction

(C) at t > 0, there is no exchange of energy between the inductor and capacitor

(D) at any time t > 0, instantaneous current in the circuit may $V\sqrt{\frac{C}{L}}$

EM0375

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current\EM\E.AC\Eng\02_EM\E.AC\02_Exercises.p65

87. If the total charge stored in the LC circuit is Q_0, then for $t \geq 0$ **[JEE 2006]**

(A) the charge on the capacitor is $Q = Q_0 \cos\left(\frac{\pi}{2} + \frac{t}{\sqrt{LC}}\right)$

(B) the charge on the capacitor is $Q = Q_0 \cos\left(\frac{\pi}{2} - \frac{t}{\sqrt{LC}}\right)$

(C) the charge on the capacitor is $Q = -LC\frac{d^2Q}{dt^2}$

(D) the charge on the capacitor is $Q = -\frac{1}{\sqrt{LC}}\frac{d^2Q}{dt^2}$

EM0375

MATRIX MATCH TYPE QUESTIONS

B. Motional EMF

88. Match the following Columns **[JEE 2006]**

	Column-I		**Column-II**
(A)	Dielectric ring uniformly charged	(P)	Time independent electrostatic field out of system
(B)	Dielectric ring uniformly charged rotating with angular velocity .	(Q)	Magnetic field
(C)	Constant current i_0 in ring	(R)	Induced electric field
(D)	Current $i = i_0 \cos \omega t$ in ring	(S)	Magnetic moment

EM0376

89. Column-I gives certain situations in which a straight metallic wire of resistance R is used and Column-II gives some resulting effects. Match the statements in Column-I with the statements in Column-II. **[JEE 2007]**

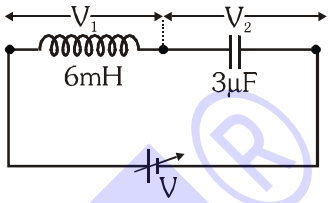
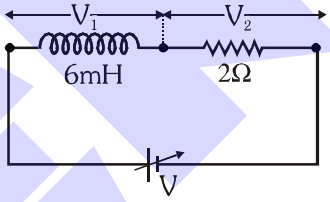
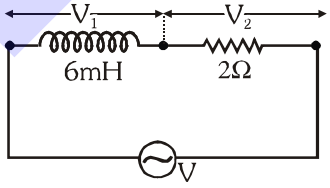
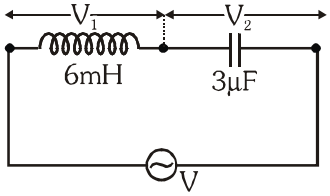
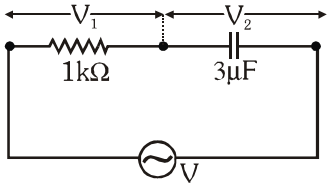
	Column-I		**Column-II**
(A)	A charged capacitor is connected to the ends of the wire	(P)	A constant current flows through the wire
(B)	The wire is moved perpendicular to its length with constant velocity in a uniform magnetic field perpendicular to the plane of motion.	(Q)	Thermal energy is generated in the wire
(C)	The wire is placed in a constant electric field that has a direction along the length of the wire	(R)	A constant potential difference develops between the ends of the wire
(D)	A battery of constant emf is connected to the ends of the wire	(S)	Charges of constant magnitude appear at the ends of the wire

EM0377

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G. AC circuits

90. You are given many resistances, capacitors and inductors. These are connected to a variable DC voltage source (the first two circuits) or an AC voltage source of 50 Hz frequency (the next three circuits) in different ways as shown in **Column-II**. When a current I (steady state for DC or rms for AC) flows through the circuit, the corresponding voltage V_1 and V_2 (indicated in circuits) are related as shown in **Column-I**. Match the two **[JEE 2010]**

Column-I		**Column-II**
(A)	$I \neq 0$, V_1 is proportional to I	(P) 
(B)	$I \neq 0$, $V_2 > V_1$	(Q) 
(C)	$V_1=0$, $V_2=V$	(R) 
(D)	$I \neq 0$, V_2 is proportional to I	(S) 
		(T) 

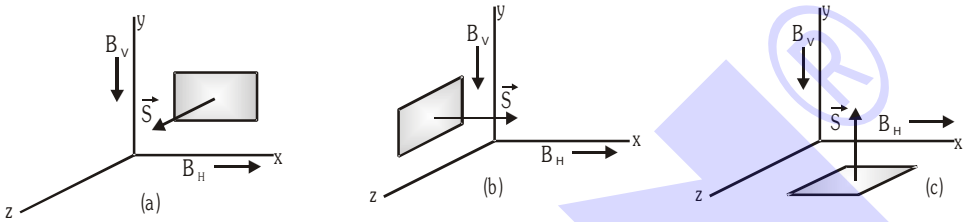
EM0196

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EXERCISE (S)

A. *Faraday law and Lenz law*

1. At a given plane, horizontal and vertical components of earth's magnetic field B_H and B_V are along x and y axes respectively as shown in figure. What is the total flux of earth's magnetic field associated with an area S, if the area S is in (a) x-y plane (b) y-z plane and (c) z-x plane ?



EM0378

2. It is desired to measure the magnitude of field between the poles of a powerful loud speaker magnet. A small flat search coil of area 2 cm^2 with 25 closely wound turns, is positioned normal to the field direction, and then quickly snatched out of the field region. Equivalently, one can give it a quick 90° turn to bring its plane parallel to the field direction. The total charge flown In the coil (measured by a ballistic galvanometer connected to coil) is 7.5 mC. The combined resistance of the coil and the galvanometer is 0.50 Ω. Estimate the field strength of magnet. **(NCERT)**

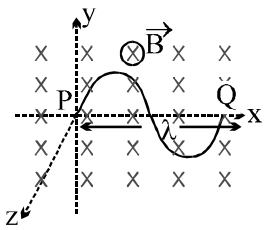
EM0006

3. An air -cored solenoid of length 30 cm. area of cross-section 25 cm^2 and number of turns 500, carries a current of 2.5 A. The current is suddenly switched off in a brief time of 10^{-3}s. How much is the average back emf induced across the ends of the open switch in the circuit? Ignore the variation in magnetic field near the ends of the solenoid. **(NCERT)**

EM0008

B. *Motional EMF*

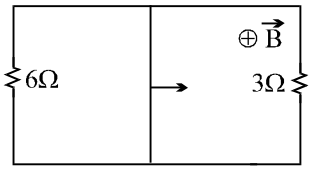
4. A wire forming one cycle of sine curve is moved in x-y plane with velocity $\vec{V} = V_x\hat{i} + V_y\hat{j}$. There exist a magnetic field $\vec{B} = -B_0\hat{k}$. Find the motional emf develop across the ends PQ of wire.



EM0001

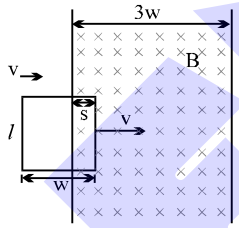
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5. A rectangular loop with a sliding connector of length $l = 1.0$ m is situated in a uniform magnetic field B = 2T perpendicular to the plane of loop. Resistance of connector is r = 2Ω. Two resistances of 6Ω and 3Ω are connected as shown in figure. Find the external force required to keep the connector moving with a constant velocity v = 2m/s.



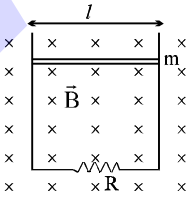
EM0003

6. A rectangular loop of dimensions l & w and resistance R moves with constant velocity V to the right as shown in the figure. It continues to move with same speed through a region containing a uniform magnetic field B directed into the plane of the paper & extending a distance 3 w. Sketch the flux, induced emf & external force acting on the loop as a function of the distance.



EM0004

7. A horizontal wire is free to slide on the vertical rails of a conducting frame as shown in figure. The wire has a mass m and length l and the resistance of the circuit is R. If a uniform magnetic field B is directed perpendicular to the frame, then find the terminal speed of the wire as it falls under the force of gravity.



EM0005

8. An aircraft with a wing span of 40 m flies with a speed of 1080 kmh^{-1} in the eastward direction at a constant altitude in the northern hemisphere, where the vertical component of earth's magnetic field is 1.75×10^{-5} T. Find the emf that develops between the tips of the wings.

EM0379

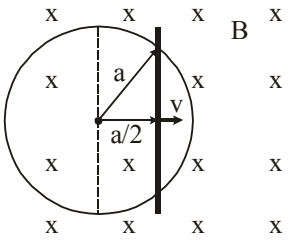
9. A coil of 160 turns of cross-sectional area 250 cm^2 rotates at an angular velocity of 300 radian/sec about an axis parallel to the plane of the coil in a uniform magnetic field of 0.6 Wb/m^2. What is the maximum emf induced in the coil ? If the coil is connected to a resistance of 2 ohm, what is the maximum torque that has to be delivered to maintain its motion.

EM0380

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10. Figure shows a uniform circular loop of radius 'a' having specific resistance ρ placed in a uniform magnetic field B perpendicular to plane of figure. A uniform rod of length 2a & resistance R moves with a velocity v as shown. Find the current in the rod when it has moved a distance $\frac{a}{2}$ from the centre of circular loop.

[Given B = √3, a = 2, v = 3, $\rho = \frac{9}{4\pi}$, R = 2/√3 all in SI units]



EM0381

11. Two straight conducting rod are connected by flexible conducting wires to form a rectangular outline (figure i). The sides of the rectangle are a = 30 cm and b = 50 cm. The circuit is placed in a uniform magnetic field with induction $B = 5 \times 10^{-3}$ T directed perpendicular to the plane. What is the charge Δq (in μC) that flows along the contour, if you turn it by 180° symmetrically as shown without the wires touching each other (figure ii)? Loop resistance R = 1 ohm. Take self-inductance of the system to be L = 1 H. Fill $\frac{\Delta q}{150}$ in OMR sheet.

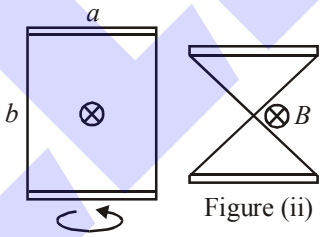
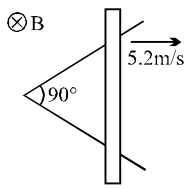


Figure (ii)

Figure (i)

EM0382

12. Two straight conducting rails form a right angle where their ends are joined. A conducting bar contact with the rails starts at vertex at the time t = 0 & moves symmetrically with a constant velocity of 5.2 m/s to the right as shown in figure. A 0.35 T magnetic field points out of the page. Calculate:
(i) The flux through the triangle by the rails & bar at t = 3.0 s.
(ii) The emf around the triangle at that time.
(iii) In what manner does the emf around the triangle vary with time .



EM0027

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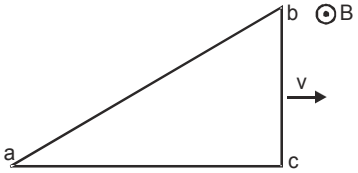
13. A right-angled triangle abc, made from a metallic wire, moves at a uniform speed v in its plane as shown in figure. A uniform magnetic field B exists in the perpendicular direction. Find the emf induced

[a] In the loop abc,

[b] In the segment bc,

[c] In the segment ac and

[d] In the segment ab



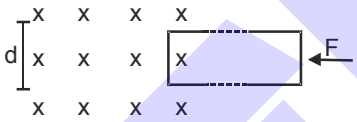
EM0383

14. The rectangular wire-frame, shown in figure, has a width d, mass m, resistance R and a large length. A uniform magnetic field B exists to the left of the frame. A constant force F starts pushing the frame into t the magnetic field at t = 0.

[a] Find the acceleration of the frame when its speed has increased to v.

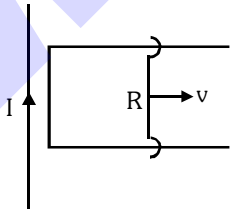
[b] Show that after some time the frame will move with a constant velocity till the whole frame enters into the magnetic field. Find the velocity v_0.

[c] Show that the velocity at time t is given by velocity at time t is given by $v = v_0\left(1 - e^{-\frac{Ft}{mv_0}}\right)$



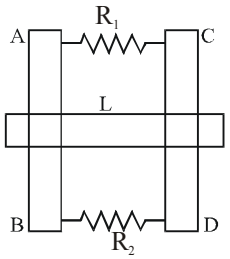
EM0384

15. A long straight wire carrying a current I and a U-shaped conductor with sliding connector are located in the same plane as shown in Fig. The connector of length ℓ and resistance R slides to the right with a constant velocity v. Find the current induced in the loop as a function of separation r between the connector and the straight wire. The resistance of the U-shaped conductor and the self-inductance of the loop are assumed to be negligible.



EM0385

16. Two parallel vertical metallic rails AB & CD are separated by 1 m. They are connected at the two ends by resistance R_1 & R_2 as shown in the figure. A horizontally metallic bar L of mass 0.2 kg slides without friction, vertically down the rails under the action of gravity. There is a uniform horizontal magnetic field of 0.6T perpendicular to the plane of the rails, it is observed that when the terminal velocity is attained, the power dissipated in R_1 & R_2 are 0.76 W & 1.2 W respectively. Find the terminal velocity of bar L & value R_1 & R_2.



EM0028

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17. A metal rod of resistance 20Ω is fixed along a diameter of a conducting ring of radius 0.1 m and lies on x-y plane. There is a magnetic field $\vec{B} = (50T)\hat{k}$. The ring rotates with an angular velocity ω = 20 rad/sec about its axis. An external resistance of 10Ω is connected across the centre of the ring and rim. Find the current through external resistance.

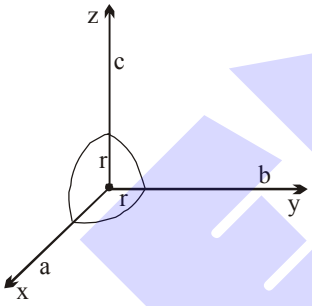
EM0030

C. Induced electric field

18. A wire is bent into 3 circular segments of radius r = 10 cm as shown in figure. Each segment is a quadrant of a circle, ab lying in the xy plane, bc lying in the yz plane & ca lying in the zx plane.

(i) if a magnetic field B points in the positive x direction, what is the magnitude of the emf developed in the wire, when B increases at the rate of 3 mT/s ?

(ii) what is the direction of the current in the segment bc.

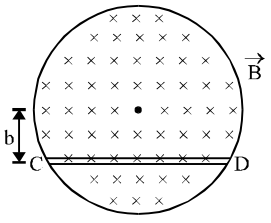


EM0002

19. There exists a uniform cylindrically symmetric magnetic field directed along the axis of a cylinder but varying with time as B = kt. If an electron is released from rest in this field at a distance of 'r' from the axis of cylinder, its acceleration, just after it is released would be (e and m are the electronic charge and mass respectively)

EM0007

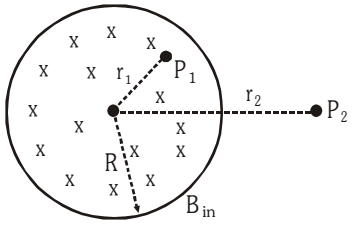
20. A uniform magnetic field $\vec{B}$ fills a cylindrical volumes of radius R. A metal rod CD of length l is placed inside the cylinder along a chord of the circular cross-section as shown in the figure. If the magnitude of magnetic field increases in the direction of field at a constant rate dB/dt, find the magnitude and direction of the EMF induced in the rod.



EM0009

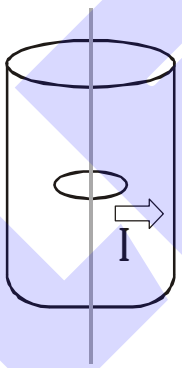
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21. For the situation described in figure the magnetic field changes with time according to
$B = (2.00\ t^3 - 4.00\ t^2 + 0.8)$ T and $r_2 = 2R = 5.0$ cm
(a) Calculate the force on an electron located at P_2 at t = 2.00 s
(b) What are the magnetude and direction of the electric field at P_1 when t = 3.00 s and $r_1 = 0.02$m.



EM0386

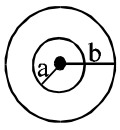
22. A long circular tube of length 10 m and radius 0.3 m carries a current I along its curved surface as shown. A wire-loop of resistance 0.005 ohm and of radius 0.1 m is placed inside the tube with its axis coinciding with the axis of the tube. The current varies as $I = I_0 \cos(300t)$ where I_0 is constant. If the magnetic moment of the loop is $N\mu_0 I_0 \sin(300t)$, then 'N' is **[JEE 2011]**



EM0199

D. *Self inductance and mutual inductance*

23. Two concentric and coplanar circular coils have radii a and b(>>a)as shown in figure. Resistance of the inner coil is R. Current in the outer coil is increased from 0 to i, then find the total charge circulating the inner coil.



EM0011

24. Find the dimension of the quantity $\frac{L}{RCV}$, where symbols have usual meaning.

EM0012

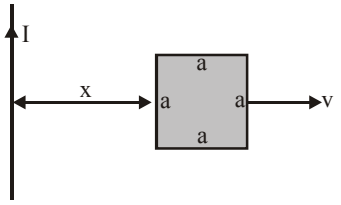
25. A pair of adjacent coils has a mutual inductance of 1.5 H. If the current in one coil changes from 0 to 20 A in 0.5 s, what is the change of flux linkage with the other coil ? **(NCERT)**

EM0018

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26. (a) Obtain an expression for the mutual inductance between a long straight wire and a square loop of side a as shown In figure.

(b) Now assume that the straight wire carries a current of 50 A and the loop is moved to the right with a constant velocity, v = 10 m/s. Calculate the induced emf in the loop at the instant when x = 0.2m. Take a = 0.1 m and assume that the loop has a large resistance. **(NCERT)**



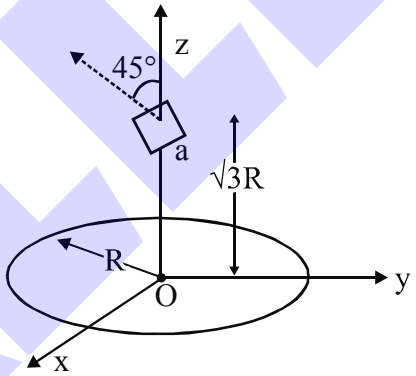
EM0019

27. Two concentric coplanar circular loops made of wire, resistance per unit length 10^{-4} Ωm^{-1}, have diameters 0.2 m and 2 m. A time-varying potential difference (4 + 2.5 t) volt is applied to the larger loop. Calculate the current in the smaller loop.

EM0387

28. A circular wire loop of radius R is placed in the x-y plane centred at the origin O. A square loop of side a (a<<R) having two turns is placed with its centre at z = $\sqrt{3}$ R along the axis of the circular wire loop, as shown in figure. The plane of the square loop makes an angle of 45° with respect to the z-axis. If the mutual inductance between the loops is given by $\frac{\mu_0 a^2}{2^{p/2} R}$, then the value of p is :-

[JEE 2012]



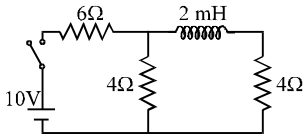
EM0201

29. Two coils, 1 & 2, have a mutual inductance = M and resistances R each. A current flows in coil 1, which varies with time as: $I_1 = kt^2$, where k is a constant and 't' is time. Find the total charge that has flown through coil 2, between t = 0 and t = T.

EM0045

E. R-L circuit

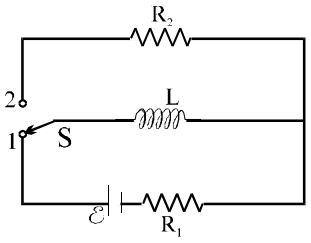
30. In the given circuit, find the ratio of i_1 to i_2 where i_1 is the initial (at t = 0) current and i_2 is steady state (at t = ∞) current through the battery.



EM0010

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31. In the circuit shown, initially the switch is in position 1 for a long time. Then the switch is shifted to position 2 for a long time. Find the total heat produced in R_2.

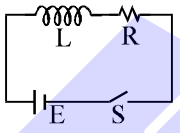


EM0013

32. An emf of 15 volt is applied in a circuit containing 5 H inductance and 10 Ω resistance. Find the ratio of the currents at time $t = \infty$ and $t = 1$ second.

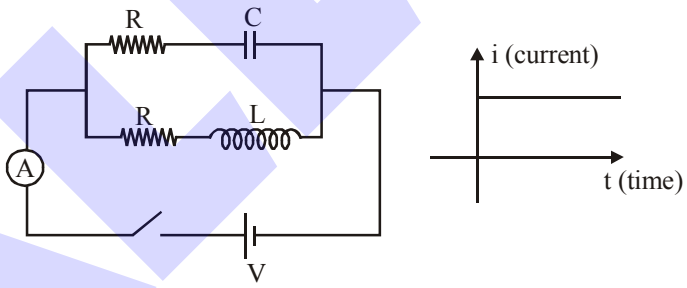
EM0014

33. In the circuit shown in figure switch S is closed at time $t = 0$. Find the charge which passes through the battery in one time constant.



EM0015

34. In the circuit shown, switch is closed at $t = 0$. The graph showing variation of current passing through ammeter with time is given below. If $C = 4\mu F$ and $L = 0.1$ mH then find the value of R (in Ω).



EM0388

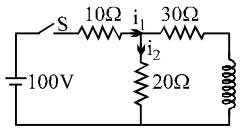
35. Find the values of i_1 and i_2

(i) immediately after the switch S is closed.

(ii) long time later, with S closed.

(iii) immediately after S is open.

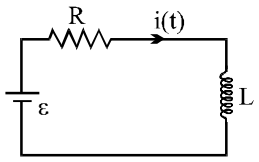
(iv) long time after S is opened.



EM0039

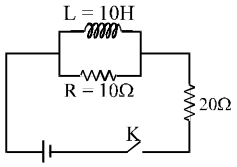
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36. Suppose the emf of the battery in the circuit shown varies with time t so the current is given by i(t) = 3 + 5t, where i is in amperes & t is in seconds. Take R = 4Ω, L = 6H & find an expression for the battery emf as function of time.



EM0040

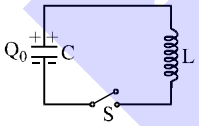
37. Two resistors of 10Ω and 20Ω and an ideal inductor of 10H are connected to a 2V battery as shown. The key K is shorted at time t = 0. Find the initial (t = 0) and final ($t \to \infty$) currents through battery.



EM0044

F. L-C oscillation

38. A capacitor C with a charge Q_0 is connected across an inductor through a switch S. If at t = 0, the switch is closed, then find the instantaneous charge q on the upper plate of capacitor.



EM0016

39. An inductor of inductance 2.0mH,is connected across a charged capacitor of capacitance 5.0μF and the resulting LC circuit is set oscillating at its natural frequency. Let Q denote the instantaneous charge on the capacitor, and I the current in the circuit .It is found that the maximum value of Q is 200μC.

(a) when Q = 100μC, what is the value of $|dI/dt|$?

(b) when Q = 200 μC, what is the value of I ?

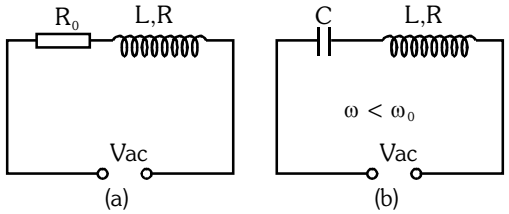
(c) Find the maximum value of I.

(d) when I is equal to one half its maximum value, what is the value of $|Q|$

EM0017

G. AC circuits

40. Draw the approximate voltage vector diagrams in the electric circuits shown in Fig. a, b. The external voltage V is assumed to be alternating harmonically with frequency ω.



EM0020

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41. A current of 4 A flows in a coil when connected to a 12 V dc source. If the same coil is connected to a 12V, 50 rad/s ac source a current of 2.4 A flows in the circuit. Determine the inductance of the coil. Also find the power developed in the circuit if a 2500 μF capacitor is connected in series with the coil.

EM0021

42. An LCR series circuit with 100Ω resistance is connected to an ac source of 200 V and angular frequency 300 rad/s. When only the capacitance is removed, the current lags behind the voltage by 60°. When only the inductance is removed, the current leads the voltage by 60°. Calculate the current and the power dissipated in the LCR circuit.

EM0022

43. A series LCR circuit containing a resistance of 120Ω has angular resonance frequency 4×10^5 rad s^{-1}. At resonance the voltages across resistance and inductance are 60 V and 40 V respectively. Find the values of L and C. At what frequency the current in the circuit lags the voltage by 45°?

EM0023

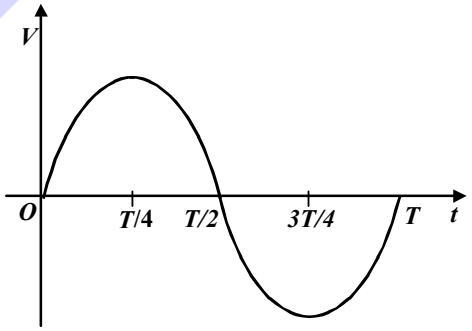
44. Find the value of an inductance which should be connected in series with a capacitor of 5μF, a resistance of 10Ω and an ac source of 50 Hz so that the power factor of the circuit is unity.

EM0024

45. In an L-R series A.C circuit the potential difference across an inductance and resistance joined in series are respectively 12 V and 16V. Find the total potential difference across the circuit.

EM0025

46. In an LR series circuit, a sinusoidal voltage $V = V_o \sin \omega t$ is applied. It is given that L = 35 mH, R = 11 Ω, $V_{rms} = 220$ V, $\frac{\omega}{2\pi} = 50$ Hz and $\pi = 22/7$. Find the amplitude of current in the steady state and obtain the phase difference between the current and the voltage. Also plot the variation of current for one cycle on the given graph. **[JEE 2004]**



EM0026

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current EM1 & AC\Eng\02_EM1 & AC\02_Exercises.p65

47. A small town with a demand of 800 kW of electric power at 220 V is situated 15 km away from an electric power plant generating power at 440 V. The resistance of the two wire line carrying power is 0.5 Ω per km. The town gets from the line through a 4000 – 220 V step down transformer at a sub-station in the town.

(a) Estimate the line power loss in the form of heat.

(b) How much power must be plant supply, assuming there is a negligible power loss due to leakage?

(c) Characterise the step up transformer at the plant.

EM0389

48. If V = 100 sin 100 t volt and I = 100 sin $(100\,t + \frac{\pi}{3})$ mA for an A.C. circuit then find out

(a) phase difference between V and I

(b) total impedance, reactance, resistance

(c) power factor and power dissipated

(d) components contains by circuits

EM0390

49. A circuit consisting of an inductance and a resistacne joined to a 200 volt supply (A.C.). It draws a current of 10 ampere. If the power used in the circuit is 1500 watt. Calculate the wattless current.

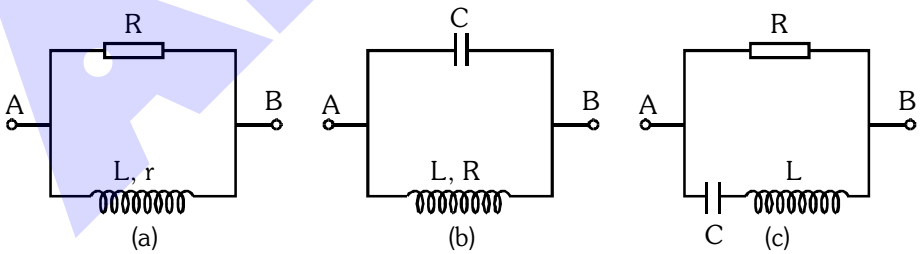
EM0391

50. A series R-C combination is connected to an AC voltage of angular frequency ω = 500 radian/s. If the impedance of the R-C circuit is $R\sqrt{1.25}$, the time constant (in millisecond) of the circuit is :-

[JEE 2011]

EM0200

51. Draw the approximate vector diagrams of currents in the circuits shown in Fig. The voltage applied across the points A and B is assumed to be sinusoidal; the parameters of each circuit are so chosen that the total current I_0 lags in phase behind the external voltage by an angle ϕ.



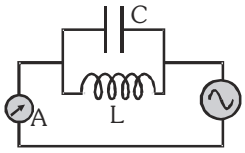
EM0047

52. A coil, a capacitor and an A.C. source of rms voltage 24 V are connected in series, By varying the frequency of the source, a maximum rms current 6 A is observed, If this coil is connected to a bettery of emf 12 V, and internal resistance 4Ω , then calculate the current through the coil.

EM0392

node06\B0BC-BD\Kota\JEE(Advanced)\Enfinity\Phy Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

53. In a L–C circuit parallel combination of inductance of 0.01 H and a capacitor of 1 μF is connected to a variable frequency alternating current source as shown in figure. Draw a rough sketch of the current variation as the frequency is changed from 1kHz to 3kHz.



EM0393

54. A choke coil and a resistance are connected in series in an a.c circuit and a potential of 130 volt is applied to the circuit. If the potential across the resistance is 50 V. What would be the potential difference across the choke coil.

EM0394

55. Calculate the resistance or inductance required to operate a lamp (60V, 10W) from a source of (100 V, 50 Hz)

EM0395

56. A box P and a coil Q are connected in series with an ac source of variable frequency. The emf of source is 10 V. Box P contains a capacitance of 1μF in series with a resistance of 32Ω while coil Q has a self-inductance 4.9 mH and a resistance of 68Ω series. The frequency is adjusted so that the maximum current flows in P and Q. Find the impedance of P and Q at this frequency. Also find the voltage across P and Q respectively.

EM0046

57. An oscillating circuit consisting of a coil and a capacitor connected in series is fed an alternating emf, with coil inductance being chosen to provide the maximum current in the circuit. Find the quality factor of the system, provided an n-fold increase of inductance results in an η-fold decrease of the current in the circuit.

EM0396

H. Damped and Forced oscillation

58. In a circuit with Q factor = 5000 and oscillation frequency 2×10^5 rad/s, the current amplitude decreases by a factor of e in a time inteval Δt. Find Δt (in sec).

EM0397

59. The position of a particle varies according to the law $x = A\cos(\omega t - \varphi)$ under the force $F = F_0 \cos \omega t$. What is the average power of this force?

EM0398

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects of Current EM & AC\Eng\02_EM & AC\02_Exercises.p65

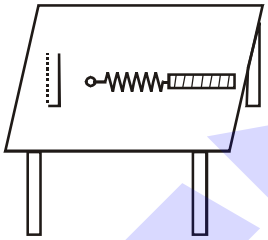
EXERCISE (JM)

1. The amplitude of adamped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10s it will decrease to a times its original magnitude, where a equals: **[JEE Main-2013]**

(1) 0.81 (2) 0.729 (3) 0.6 (4) 0.7

EM0183

2. A metallic rod of length 'l' is tied to a string of length 2l and made to rotate with angular speed ω on a horizontal table with one end of the string fixed. If there is a vertical magnetic field 'B' in the region, the e.m.f. induced across the ends of the rod is : **[JEE Main-2013]**



(1) $\frac{2B\omega l^2}{2}$ (2) $\frac{3B\omega l^2}{2}$ (3) $\frac{4B\omega l^2}{2}$ (4) $\frac{5B\omega l^2}{2}$

EM0184

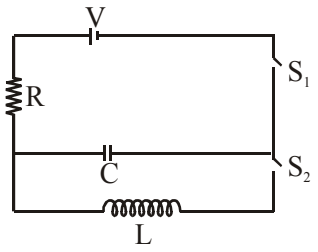
3. A circular loop of radius 0.3 cm lies parallel to a much bigger circular loop of radius 20 cm. The centre of the small loop is on the axis of the bigger loop. The distance between their centres is 15 cm. If a current of 2.0 A flows through the smaller loop, then the flux linked with bigger loop is :-

[JEE Main-2013]

(1) 9.1×10^{-11} weber (2) 6×10^{-11} weber (3) 3.3×10^{-11} weber (4) 6.6×10^{-9} weber

EM0185

4. In an LCR circuit as shown below both switches are open initially. Now switch S_1 is closed, S_2 kept open, (q is charge on the capacitor and $\tau = RC$ is Capacitive time constant). Which of the following statement is correct? **[JEE Main-2013]**



(1) Work done by the battery is half of the energy dissipated in the resistor

(2) At $t = \tau$, $q = CV/2$

(3) At $t = 2\tau$, $q = CV(1-e^{-2})$

(4) At $t = \frac{\tau}{2}$, $q = CV(1-e^{-1})$

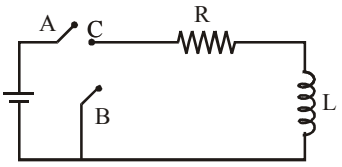
EM0186

node06\B0BC-BD\Kota\JEE(Advanced)\Enfinity\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

5. In the circuit shown here, the point 'C' is kept connected to point 'A' till the current flowing through the circuit becomes constant. Afterward, suddenly, point 'C' is disconnected from point 'A' and connected to point 'B' at time t = 0. Ratio of the voltage across resistance and the inductor at t = L/R will be equal to : **[JEE Main-2014]**

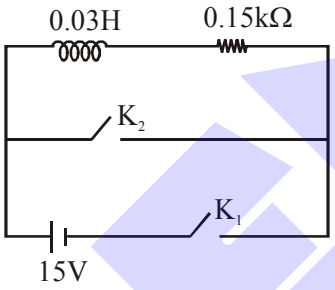
(1) -1 (2) $\frac{1-e}{e}$

(3) $\frac{e}{1-e}$ (4) 1



EM0187

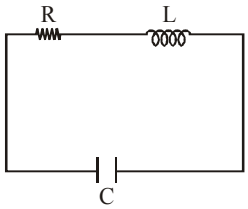
6. An inductor (L = 0.03 H) and a resistor (R = 0.15 kΩ) are connected in series to a battery of 15V EMF in a circuit shown below. The key K_1 has been kept closed for a long time. Then at t = 0, K_1 is opened and key K_2 is closed simultaneously. At t = 1ms, the current in the circuit will be ($e^5 \cong 150$):- **[JEE Main-2015]**

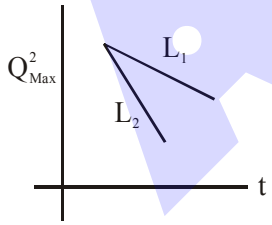


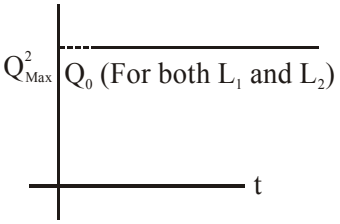
(1) 6.7 mA (2) 0.67 mA (3) 100 mA (4) 67 mA

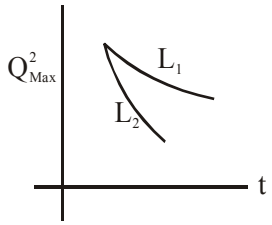
EM0188

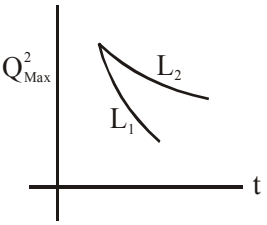
7. An LCR circuit is equivalent to a damped pendulum. In an LCR circuit the capacitor is charged to Q_0 and then connected to the L and R as shown below. If a student plots graphs of the square of maximum charge (Q^2_{Max}) on the capacitor with time (t) for two different values L_1 and L_2 ($L_1 > L_2$) of L then which of the following represents this graph correctly ? (plots are schematic and not drawn to scale) **[JEE Main-2015]**



(1)


(2)


(3)


(4)


EM0189

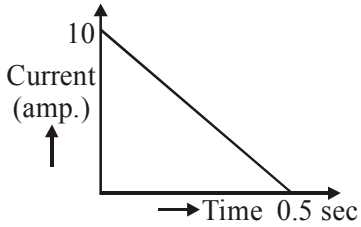
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EM1 & AC\Eng\02_EM1 & AC\02_Exercises.p65

8. An arc lamp requires a direct current of 10A at 80V to function. If it is connected to a 220V (rms), 50Hz AC supply, the series inductor needed for it to work is close to :- **[JEE Main-2016]**

(1) 0.065 H (2) 80 H (3) 0.08 H (4) 0.044 H

EM0190

9. In a coil of resistance 100 Ω, a current is induced by changing the magnetic flux through it as shown in the figure. The magnitude of change in flux through the coil is :- **[JEE Main-2017]**



(1) 250 Wb (2) 275 Wb (3) 200 Wb (4) 225 Wb

EM0191

10. For an RLC circuit driven with voltage of amplitude v_m and frequency $w_0 = \frac{1}{\sqrt{LC}}$ the current exhibits resonance. The quality factor, Q is given by :- **[JEE Main-2018]**

(1) $\frac{\omega_0 R}{L}$ (2) $\frac{R}{(\omega_0 C)}$ (3) $\frac{CR}{\omega_0}$ (4) $\frac{\omega_0 L}{R}$

EM0192

11. In an a. c. circuit, the instantaneous e.m.f. and current are given by

$e = 100 \sin 30\, t$

$i = 20 \sin\left(30t - \frac{\pi}{4}\right)$

In one cycle of a.c., the average power consumed by the circuit and the wattless current are, respectively. **[JEE Main-2018]**

(1) $\frac{1000}{\sqrt{2}}$, 10 (2) $\frac{50}{\sqrt{2}}$, 0 (3) 50, 0 (4) 50, 10

EM0193

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

12. A series AC circuit containing an inductor (20 mH), a capacitor (120 μF) and a resistor (60Ω) is driven by an AC source of 24V/50Hz. The energy dissipated in the circuit in 60 s is : **[JEE Main-2019 Jan.]**

(1) 2.26×10^3 J (2) 3.39×10^3 J (3) 5.65×10^2 J (4) 5.17×10^2 J

EM0239

13. A power transmission line feeds input power at 2300 V to a step down transformer with its primary windings having 4000 turns. The output power is delivered at 230 V bv the transformer. If the current in the primary of the transformer is 5A and its efficiency is 90%, the output current would be : **[JEE Main-2019 Jan.]**

(1) 25 A (2) 50 A (3) 35 A (4) 45 A

EM0240

node06\B0BC-BD\Kota\JEE(Advanced)\Enfrusve\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

14. A copper wire is wound on a wooden frame, whose shape is that of an equilateral triangle. If the linear dimension of each side of the frame is increased by a factor of 3, keeping the number of turns of the coil per unit length of the frame the same, then the self inductance of the coil :

[JEE Main-2019 Jan.]

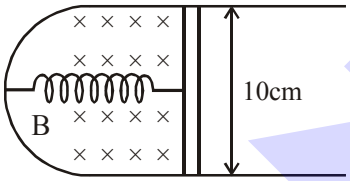
(1) Decreases by a factor of $9\sqrt{3}$ (2) Increases by a factor of 3

(3) Decreases by a factor of 9 (4) Increases by a factor of 27

EM0241

15. A thin strip 10 cm long is on a U shaped wire of negligible resistance and it is connected to a spring of spring constant 0.5 Nm^{-1} (see figure). The assembly is kept in a uniform magnetic field of 0.1 T. If the strip is pulled from its equilibrium position and released, the number of oscillation it performs before its amplitude decreases by a factor of e is N. If the mass of the strip is 50 grams, its resistance 10Ω and air drag negligible, N will be close to :

[JEE Main-2019 April]



(1) 50000 (2) 5000 (3) 10000 (4) 1000

EM0242

16. An alternating voltage v(t) = 220 sin 100 πt volt is applied to a purely resistance load of 50 Ω. The time taken for the current to rise from half of the peak value to the peak value is :

[JEE Main-2019 April]

(1) 2.2 ms (2) 5 ms (3) 3.3 ms (4) 7.2 ms

EM0243

17. The total number of turns and cross-section area in a solenoid is fixed. However, its length L is varied by adjusting the separation between windings. The inductance of solenoid will be proportional to :

[JEE Main-2019 April]

(1) $1/L^2$ (2)1/L (3) L (4) L^2

EM0244

18. The displacement of a damped harmonic oscillator is given by
$x(t) = e^{-0.1t} \cos(10\pi t + \phi)$. Here t is in seconds.
The time taken for its amplitude of vibration to drop to half of its initial value is close to :

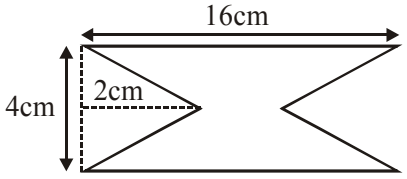
[JEE Main-2019 April]

(1) 13 s (2) 7 s (3) 27 s (4) 4 s

EM0245

19. At time t = 0 magnetic field of 1000 Gauss is passing perpendicularly through the area defined by the closed loop shown in the figure. If the magnetic field reduces linearly to 500 Gauss, in the next 5s, then induced EMF in the loop is :

[JEE Main-2020_Jan.]



(1) 36 μV (2) 48 μV (3) 56 μV (4) 28 μV

EM0246

node06\B0BC-BD\Data\JEE\Advanced\Enthuse\Phy\Module\3-Magnetic Effects of Current EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

20. A uniform magnetic field B exists in a direction perpendicular to the plane of a square loop made of a metal wire. The wire has a diameter of 4 mm and a total length of 30 cm. The magnetic field changes with time at a steady rate dB/dt = 0.032 Ts^{-1}.The induced current in the loop is close to (Resistivity of the metal wire is 1.23×10^{-8} Ωm) **[JEE Main-2020_Sep.]**

(1) 0.61 A (2) 0.34 A (3) 0.43 A (4) 0.53 A

EM0247

21. A 750 Hz, 20 V (rms) source is connected to a resistance of 100 Ω, an inductance of 0.1803 H and a capacitance of 10 μF all in series. The time in which the resistance (heat capacity 2J/°C) will get heated by 10°C. (assume no loss of heat to the surroundings) is close to : **[JEE Main-2020_Sep.]**

(1) 418 s (2) 245 s (3) 348 s (4) 365 s

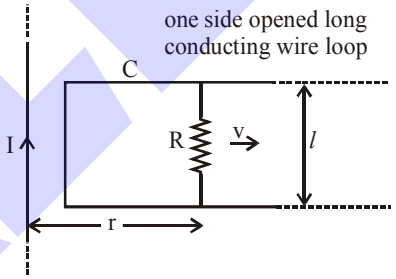
EM0248

22. A series L-R circuit is connected to a battery of emf V. If the circuit is switched on at t = 0, then the time at which the energy stored in the inductor reaches $\left(\frac{1}{n}\right)$ times of its maximum value, is : **[JEE Main-2020_Sep.]**

(1) $\frac{L}{R}\ln\left(\frac{\sqrt{n}-1}{\sqrt{n}}\right)$ (2) $\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}+1}\right)$ (3) $\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}-1}\right)$ (4) $\frac{L}{R}\ln\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)$

EM0249

23. An infinitely long straight wire carrying current I, one side opened rectangular loop and a conductor C with a sliding connector are located in the same plane, as shown in the figure. The connector has length *l* and resistance R. It slides to the right with a velocity v. The resistance of the conductor and the self inductance of the loop are negligible. The induced current in the loop, as a function of separation r, between the connector and the straight wire is : **[JEE Main-2020_Sep.]**



(1) $\frac{\mu_0}{\pi}\frac{Iv\ell}{Rr}$ (2) $\frac{\mu_0}{2\pi}\frac{Iv\ell}{Rr}$ (3) $\frac{2\mu_0}{\pi}\frac{Iv\ell}{Rr}$ (4) $\frac{\mu_0}{4\pi}\frac{Iv\ell}{Rr}$

EM0250

24. An AC circuit has R = 100 Ω, C = 2 μF and L = 80 mH, connected in series. The quality factor of the circuit is : **[JEE Main-2020_Sep.]**

(1) 0.5 (2) 2 (3) 20 (4) 400

EM0251

25. A resonance circuit having inductance and resistance 2×10^{-4} H and 6.28 Ω respectively oscillates at 10 MHz frequency. The value of quality factor of this resonator is _____. [π = 3.14]

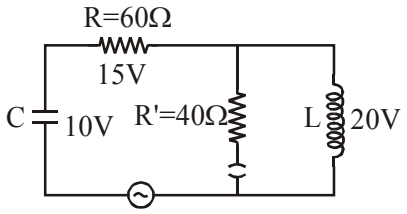
[JEE Main-2021_Feb]

EM0399

node06\B0BC-BD\Kota\JEE(Advanced)\Enfinity\Phy Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

26. The angular frequency of alternating current in a L-C-R circuit is 100 rad/s. The components connected are shown in the figure. Find the value of inductance of the coil and capacity of condenser.

[JEE Main-2021_Feb]



(1) 0.8 H and 150 μF (2) 0.8 H and 250 μF (3) 1.33 H and 250 μF (4) 1.33 H and 150 μF

EM0400

27. An alternating current is given by the equation $i = i_1 \sin\omega t + i_2 \cos\omega t$. The rms current will be

[JEE Main-2021_Feb]

(1) $\frac{1}{\sqrt{2}}\left(i_1^2 + i_2^2\right)^{\frac{1}{2}}$ (2) $\frac{1}{\sqrt{2}}\left(i_1 + i_2\right)^2$ (3) $\frac{1}{2}\left(i_1^2 + i_2^2\right)^{\frac{1}{2}}$ (4) $\frac{1}{\sqrt{2}}\left(i_1 + i_2\right)$

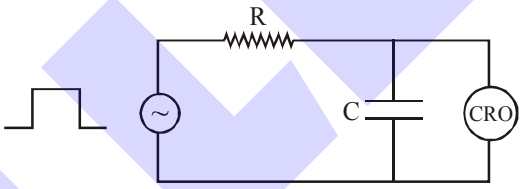
EM0401

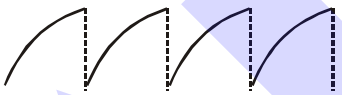
28. In a series LCR resonant circuit, the quality factor is measured as 100. If the inductance is increased by two fold and resistance is decreased by two fold, then the quality factor after this change will be_____.

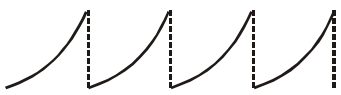
[JEE Main-2021_Feb]

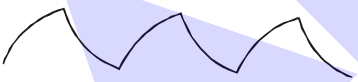
EM0402

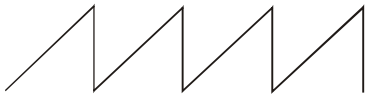
29. An RC circuit as shown in the figure is driven by a AC source generating a square wave. The output wave pattern monitored by CRO would look close to : **[JEE Main-2021_March]**



(1) 

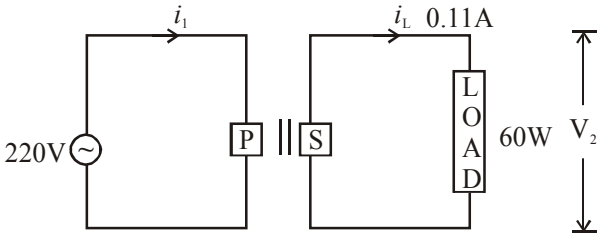
(2) 

(3) 

(4) 

EM0403

30. For the given circuit, comment on the type of transformer used : **[JEE Main-2021_March]**



(1) Auxilliary transformer (2) Auto transformer
(3) Step-up transformer (4) Step down transformer

EM0404

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EM1 & AC\Eng\02_EM1 & AC\02_Exercises.p65

31. Amplitude of a mass-spring system, which is executing simple harmonic motion decreases with time. If mass = 500g, Decay constant = 20 g/s then how much time is required for the amplitude of the system to drop to half of its initial value ? (ln 2 = 0.693) **[JEE Main-2021_March]**

(1) 34.65 s (2) 17.32 s (3) 0.034 s (4) 15.01 s

EM0405

32. Match List-I with List-II

	List-I		**List-II**
(a)	Phase difference between current and voltage in a purely resistive AC circuit	(i)	$\frac{\pi}{2}$; current leads voltage
(b)	Phase difference between current and voltage in a pure inductive AC circuit	(ii)	zero
(c)	Phase difference between current and voltage in a pure capacitive AC circuit	(iii)	$\frac{\pi}{2}$; current lags voltage
(d)	Phase difference between current and voltage in an LCR series circuit	(iv)	$\tan^{-1}\left(\frac{X_C - X_L}{R}\right)$

Choose the most appropriate answer from the options given below : **[JEE Main-2021_March]**

(1) (a)–(i),(b)–(iii),(c)–(iv),(d)–(ii) (2) (a)–(ii),(b)–(iv),(c)–(iii),(d)–(i)

(3) (a)–(ii),(b)–(iii),(c)–(iv),(d)–(i) (4) (a)–(ii),(b)–(iii),(c)–(i),(d)–(iv)

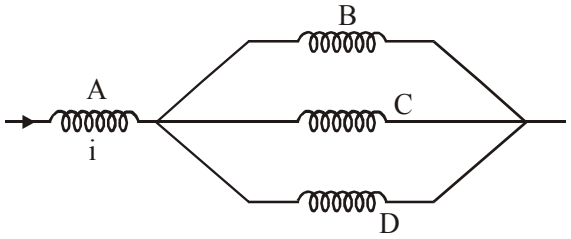
EM0406

33. Seawater at a frequency $f = 9 \times 10^2$ Hz, has permittivity $\varepsilon = 80\varepsilon_0$ and resistivity $\rho = 0.25\ \Omega$m. Imagine a parallel plate capacitor is immersed in seawater and is driven by an alternating voltage source $V(t)=V_0 \sin(2\pi ft)$. Then the conduction current density becomes 10^x times the displacement current density after time $t = \frac{1}{800}$ s. The value of x is _____ (Given : $\frac{1}{4\pi\varepsilon_0} = 9\times10^9\ Nm^2C^{-2}$)

[JEE Main-2021_March]

EM0407

34. Four identical long solenoids A, B, C and D are connected to each other as shown in the figure. If the magnetic field at the center of A is 3T, the field at the center of C would be : (Assume that the magnetic field is confined with in the volume of respective solenoid). **[JEE Main-2021_March]**



(1) 12T (2) 6T (3) 9T (4) 1T

EM0408

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35. In a series LCR resonance circuit, if we change the resistance only, from a lower to higher value:
[JEE Main-2021_March]

(1) The bandwidth of resonance circuit will increase.
(2) The resonance frequency will increase.
(3) The quality factor will increase.
(4) The quality factor and the resonance frequency will remain constant.

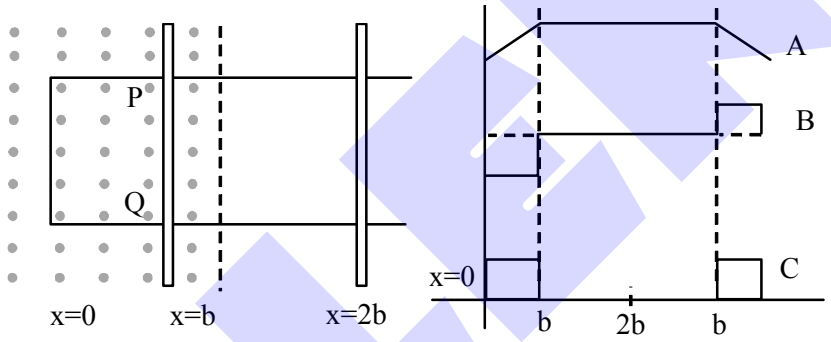
EM0409

36. AC voltage V(t) = 20 sin t of frequency 50 Hz is applied to a parallel plate capacitor. The separation between the plates is 2 mm and the area is 1 m^2. The amplitude of the oscillating displacement current for the applied AC voltage is _____. [Take $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m] **[JEE Main-2021_July]**

(1) 21.14 μA (2) 83.37 μA (3) 27.79 μA (4) 55.58 μA

EM0410

37. The arm PQ of a rectangular conductor is moving from x = 0 to x = 2b outwards and then inwards from x = 2b to x = 0 as shown in the figure. A uniform magnetic field perpendicular to the plane is acting from x = 0 to x = b. Identify the graph showing the variation of different quantities with distance:
[JEE Main-2021_July]



(1) A-Flux, B-Power dissipated, C-EMF (2) A-Power dissipated, B-Flux, C-EMF
(3) A-Flux, B-EMF, C-Power dissipated (4) A-EMF, B-Power dissipated, C-Flux

EM0411

38. Match List-I with List-II :

	List–I		List–II
(a)	$\omega L > \frac{1}{\omega C}$	(i)	Current is in phase with emf
(b)	$\omega L = \frac{1}{\omega C}$	(ii)	Current lags behind the applied emf
(c)	$\omega L < \frac{1}{\omega C}$	(iii)	Maximum current occurs
(d)	Resonant frequency	(iv)	Current leads the emf

Choose the correct answer from the options given below : **[JEE Main-2021_July]**

(1) (a) - (ii) ; (b) - (i) ; (c) - (iv) ; (d) - (iii) (2) (a) - (ii) ; (b) - (i) ; (c) - (iii) ; (d) - (iv)
(3) (a) - (iii) ; (b) - (i) ; (c) - (iv) ; (d) - (ii) (4) (a) - (iv) ; (b) - (iii) ; (c) - (ii) ; (d) - (i)

EM0412

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39. A circular conducting coil of radius 1 m is being heated by the change of magnetic field $\vec{B}$ passing perpendicular to the plane in which the coil is laid. The resistance of the coil is 2 μΩ. The magnetic field is slowly switched off such that its magnitude changes in time as

$$B = \frac{4}{\pi} \times 10^{-3} T \left(1 - \frac{t}{100}\right)$$

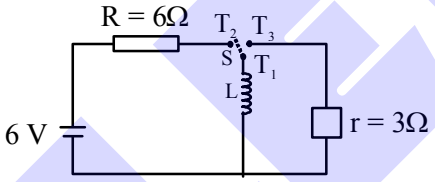
The energy dissipated by the coil before the magnetic field is switched off completely is E = __ mJ.

[JEE Main-2021_July]

EM0413

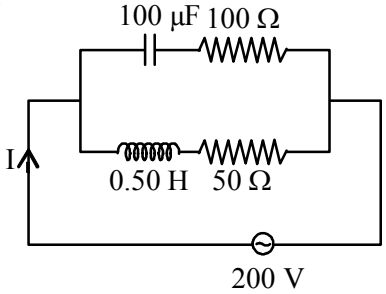
40. Consider an electrical circuit containing a two way switch 'S'. Initially S is open and then T_1 is connected to T_2. As the current in R = 6Ω attains a maximum value of steady state level, T_1 is disconnected from T_2 and immediately connected to T_3. Potential drop across r = 3Ω resistor immediately after T_1 is connected to T_3 is_____V. (Round off to the Nearest Integer)

[JEE Main-2021_July]



EM0414

41. In the given circuit the AC source has ω = 100 rad s^{-1}. Considering the inductor and capacitor to be ideal, what will be the current I flowing through the circuit? **[JEE Main-2021_Aug]**

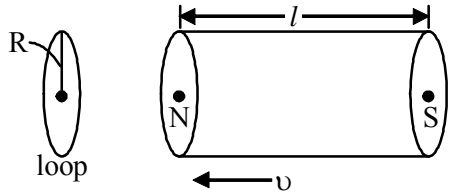


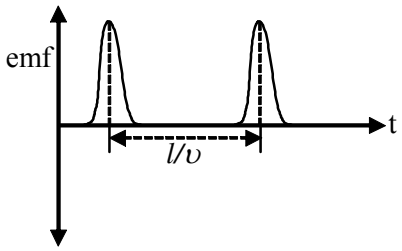
(1) 5.9 A (2) 4.24 A (3) 0.94 A (4) 6 A

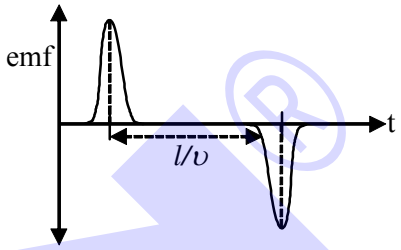
EM0415

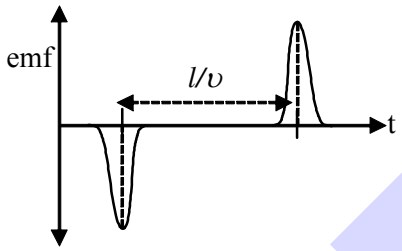
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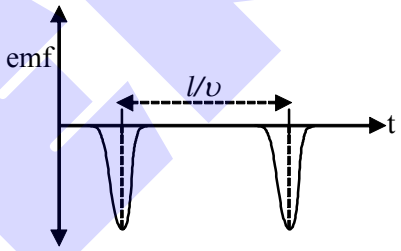
42. A bar magnet is passing through a conducting loop of radius R with velocity v. The radius of the bar magnet is such that it just passes through the loop. The induced e.m.f. in the loop can be represented by the approximate curve : **[JEE Main-2021_Aug]**



(1)


(2)


(3)


(4)


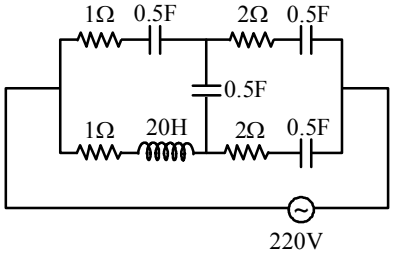
EM0416

43. A small square loop of side 'a' and one turn is placed inside a larger square loop of side b and one turn (b >> a). The two loops are coplanar with their centres coinciding. If a current I is passed in the square loop of side 'b', then the coefficient of mutual inductance between the two loops is : **[JEE Main-2021_Aug]**

(1) $\frac{\mu_0}{4\pi} 8\sqrt{2} \frac{a^2}{b}$ (2) $\frac{\mu_0}{4\pi} \frac{8\sqrt{2}}{a}$ (3) $\frac{\mu_0}{4\pi} 8\sqrt{2} \frac{b^2}{a}$ (4) $\frac{\mu_0}{4\pi} \frac{8\sqrt{2}}{b}$

EM0417

44. At very high frequencies, the effective impendance of the given circuit will be_____ Ω. **[JEE Main-2021_Aug]**



EM0418

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45. Match List-I with List-II

	List-I		**List-II**
(A)	AC generator	(I)	Detects the presence of current in the circuit
(B)	Galvanometer	(II)	Converts mechanical energy into electrical energy
(C)	Transformer	(III)	Works on the principle of resonance in AC circuit
(D)	Metal detector	(IV)	Changes an alternating voltage for smaller or greater value

Choose the **correct answer** from the options given below :- **[JEE Main-2022_June]**

(A) (A)–(II), B–(I), (C)–(IV), (D)–(III)

(B) (A)–(II), B–(I), (C)–(III), (D)–(IV)

(C) (A)–(III), B–(IV), (C)–(II), (D)–(I)

(D) (A)–(III), B–(I), (C)–(II), (D)–(IV)

EM0419

46. The current in a coil of self inductance 2.0 H is increasing according to $I = 2\sin(t^2)A$. The amount of energy spent during the period when current changes from 0 to 2A is _____ J.

[JEE Main-2022_June]

EM0420

47. Two coils of self inductance L_1 and L_2 are connected in series combination having mutual inductance of the coils as M. The equivalent self inductance of the combination will be : **[JEE Main-2022_June]**



(A) $\frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{M}$ (B) L_1+L_2+M (C) L_1+L_2+2M (D) L_1+L_2-2M

EM0421

48. An EM wave propagating in x-direction has a wavelength of 8 mm. The electric field vibrating y-direction has maximum magnitude of 60 Vm^{-1}. Choose the correct equations for electric and magnetic fields if the EM wave is propagating in vacuum : **[JEE Main-2022_June]**

(A) $E_y = 60\sin\left[\frac{\pi}{4}\times10^3\left(x-3\times10^8 t\right)\right]\hat{j}Vm^{-1}$, $B_z = 2\sin\left[\frac{\pi}{4}\times10^3\left(x-3\times10^8 t\right)\right]\hat{k}T$

(B) $E_y = 60\sin\left[\frac{\pi}{4}\times10^3\left(x-3\times10^8 t\right)\right]\hat{j}Vm^{-1}$, $B_z = 2\times10^{-7}\sin\left[\frac{\pi}{4}\times10^3\left(x-3\times10^8 t\right)\right]\hat{k}T$

(C) $E_y = 2\times10^{-7}\sin\left[\frac{\pi}{4}\times10^3\left(x-3\times10^8 t\right)\right]\hat{j}Vm^{-1}$, $B_z = 60\sin\left[\frac{\pi}{4}\times10^3\left(x-3\times10^8 t\right)\right]\hat{k}T$

(D) $E_y = 2\times10^{-7}\sin\left[\frac{\pi}{4}\times10^4\left(x-4\times10^8 t\right)\right]\hat{j}Vm^{-1}$, $B_z = 60\sin\left[\frac{\pi}{4}\times10^4\left(x-4\times10^8 t\right)\right]\hat{k}T$

EM0422

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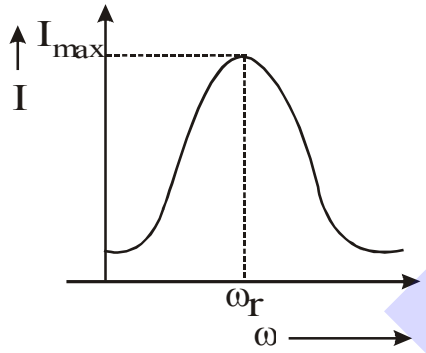
49. For a series LCR circuit, I vs ω curve is shown : **[JEE Main-2022_June]**

(a) To the left of ω_r, the circuit is mainly capacitive.

(b) To the left of ω_r, the circuit is mainly inductive.

(c) At ω_r, impedance of the circuit is equal to the resistance of the circuit.

(d) At ω_r, impedance of the circuit is 0.



Choose the most appropriate answer from the options given below :

(A) (a) and (d) only (B) (b) and (d) only

(C) (a) and (c) only (D) (b) and (c) only

EM0423

50. To light, a 50 W, 100 V lamp is connected, in series with a capacitor of capacitance $\frac{50}{\pi\sqrt{x}}\mu F$, with 200 V, 50Hz AC source. The value of x will be ____ . **[JEE Main-2022_July]**

EM0424

51. A conducting circular loop is placed in X - Y plane in presence of magnetic field $\vec{B} = \left(3t^3\hat{j} + 3t^2\hat{k}\right)$ in SI unit. If the radius of the loop is 1m, the induced emf in the loop, at time, t = 2s is nπV. The value of n is…….. **[JEE Main-2022_July]**

EM0425

52. The frequencies at which the current amplitude in an LCR series circuit becomes $\frac{1}{\sqrt{2}}$ times its maximum value, are 212 rad s^{-1} and 232 rad s^{-1}. The value of resistance in the circuit is $R = 5\Omega$. The self inductance in the circuit is _____ mH. **[JEE Main-2022_July]**

EM0426

53. A transformer operating at primary voltage 8 kV and secondary voltage 160 V serves a load of 80kW. Assuming the transformer to be ideal with purely resistive load and working on unity power factor, the loads in the primary and secondary circuit would be **[JEE Main-2022_July]**

(A) 800Ω and 1.06Ω (B) 10Ω and 500Ω

(C) 800Ω and 0.32Ω (D) 1.06Ω and 500Ω

EM0427

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EXERCISE (JA)

Paragraph for Questions 1 and 2

A point charge Q is moving in a circular orbit of radius R in the x-y plane with an angular velocity ω. This can be considered as equivalent to a loop carrying a steady current $\frac{Q\omega}{2\pi}$. A uniform magnetic field along the positive z-axis is now switched on, which increases at a constant rate from 0 to B in one second. Assume that the radius of the orbit remains constant. The application of the magnetic field induces an emf in the orbit. The induced emf is defined as the work done by an induced electric field in moving a unit positive charge around a closed loop. It is known that for an orbiting charge, the magnetic dipole moment is proportional to the angular momentum with a proportionality constant γ.

1. The change in the magnetic dipole moment associated with the orbit, at the end of the time interval of the magnetic field change is **[JEE Advance-2013]**

(A) $-\gamma BQR^2$ (B) $-\gamma\frac{BQR^2}{2}$ (C) $\gamma\frac{BQR^2}{2}$ (D) γBQR^2

EM0204

2. The magnitude of the induced electric field in the orbit at any instant of time during the time interval of the magnetic field change is

(A) $\frac{BR}{4}$ (B) $\frac{BR}{2}$ (C) BR (D) 2BR

EM0204

Paragraph for Questions 3 and 4

A thermal power plant produces electric power of 600 kW and 4000 V, which is to be transported to a place 20 km away from the power plant for consumers' usage. It can be transported either directly with a cable of large current carrying capacity or by using a combination of step-up and step-down transformers at the two ends. The drawback of the direct transmission is the large energy dissipation. In the method using transformers, the dissipation is much smaller. In this method, a step-up transformer is used at the plant side so that the current is reduced to a smaller value. At the consumers' end, a step-down transformer is used to supply power to the consumers at the specified lower voltage. It is reasonable to assume that the power cable is purely resistive and the transformers are ideal with a power factor unity. All the currents and voltages mentioned are rms values. **[JEE Advance-2013]**

3. In the method using the transformers, assume that the ratio of the number of turns in the primary to that in the secondary in the step-up transformer is 1 : 10. If the power to the consumers has to be supplied at 200 V, the ratio of the number of turns in the primary to that in the secondary in the step-down transformer is

(A) 200 : 1 (B) 150 : 1 (C) 100 : 1 (D) 50 : 1

EM0205

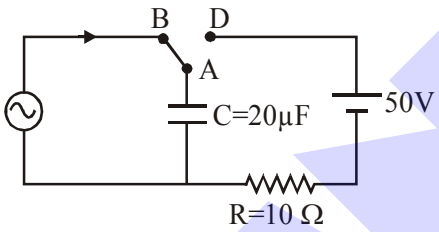
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4. If the direct transmission method with a cable of resistance 0.4 Ω km^{-1} is used, the power dissipation (in %) during transmission is

(A) 20 (B) 30 (C) 40 (D) 50

EM0205

5. At time t = 0, terminal A in the circuit shown in the figure is connected to B by a key and an alternating current $I(t) = I_0\cos(\omega t)$, with I_0 = 1A and ω = 500 rad s^{-1} starts flowing in it with the initial direction shown in the figure. At $t = \frac{7\pi}{6\omega}$, the key is switched from B to D. Now onwards only A and D are connected. A total charge Q flows from the battery to charge the capacitor fully. If C = 20 μF, R = 10 Ω and the battery is ideal with emf of 50 V, identify the correct statement (s). **[JEE Advance-2014]**



(A) Magnitude of the maximum charge on the capacitor before $t = \frac{7\pi}{6\omega}$ is 1×10^{-3}C.

(B) The current in the left part of the circuit just before $t = \frac{7\pi}{6\omega}$ is clockwise.

(C) Immediately after A is connected to D, the current in R is 10A

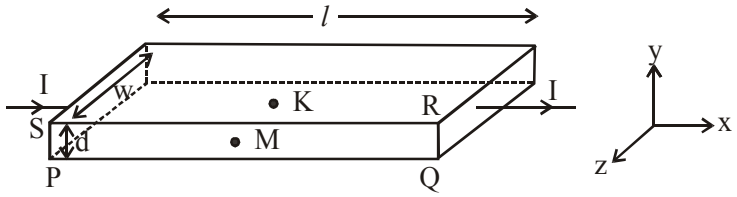
(D) Q = 2×10^{-3} C

EM0206

Paragraph for Question No. 6 and 7

In a thin rectangular metallic strip a constant current I flows along the positive x-direction, as shown in the figure. The length, width the thickness of the strip are *l*, w and d, respectively.

A uniform magnetic field $\vec{B}$ is applied on the strip along the positive y-direction. Due to this, the charge carriers experience a net deflection along the z-direction. This results in accumulation of charge carriers on the surface PQRS and appearance of equal and opposite charges on the face opposite to PQRS. A potential difference along the z-direction is thus developed. Charge accumulation continues until the magnetic force is balanced by the electric force. The current is assumed to be uniformly distributed on the cross section of the strip and carried by electrons. **[JEE Advance-2015]**



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6. Consider two different metallic strips (1 and 2) of the same material. Their lengths are the same, widths are w_1 and w_2 and thicknesses are d_1 and d_2, respectively. Two points K and M are symmetrically located on the opposite faces parallel to the x-y plane (see figure). V_1 and V_2 are the potential differences between K and M in strips 1 and 2, respectively. Then, for a given current I flowing through them in a given magnetic field strength B, the correct statement(s) is(are)

(A) If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = 2V_1$

(B) If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = V_1$

(C) If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = 2V_1$

(D) If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = V_1$

EM0207

7. Consider two different metallic strips (1 and 2) of same dimensions (length *l*, width w and thickness d) with carrier densities n_1 and n_2, respectively. Strip 1 is placed in magnetic field B_1 and strip 2 is placed in magnetic field B_2, both along positive y-direction. Then V_1 and V_2 are the potential differences developed between K and M in strips 1 and 2, respectively. Assuming that the current I is the same for both the strips, the correct option(s) is(are)

(A) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = 2V_1$

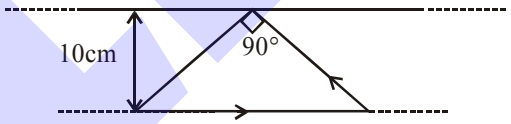
(B) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = V_1$

(C) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = 0.5V_1$

(D) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = V_1$

EM0207

8. A conducting loop in the shape of right angled isosceles triangle of height 10 cm is kept such that the 90° vertex is very close to an infinitely long conducting wire (see the figure). The wire is electrically insulated from the loop. The hypotenuse of the triangle is parallel to the wire. The current in the triangular loop is in counterclockwise direction and increased at constant rate of 10 A s^{-1}. Which of the following statement(s) is(are) true? **[JEE Advance-2016]**



(A) The induced current in the wire is in opposite direction to the current along the hypotenuse.

(B) There is a repulsive force between the wire and the loop

(C) If the loop is rotated at a constant angular speed about the wire, an additional emf of $\left(\frac{\mu_0}{\pi}\right)$ volt is induced in the wire

(D) The magnitude of induced emf in the wire is $\left(\frac{\mu_0}{\pi}\right)$ volt.

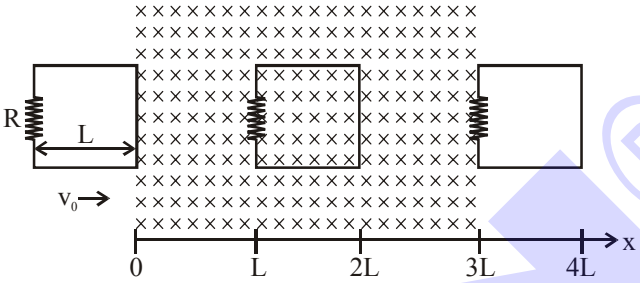
EM0208

9. Two inductors L_1 (inductance 1 mH, internal resistance 3 Ω) and L_2 (inductance 2mH, internal resistance 4Ω), and a resistor R (resistance 12 Ω) are all connected in parallel across a 5V battery. The circuit is switched on at time t = 0. The ratio of the maximum to the minimum current (I_{max}/I_{min}) drawn from the battery is. **[JEE Advance-2016]**

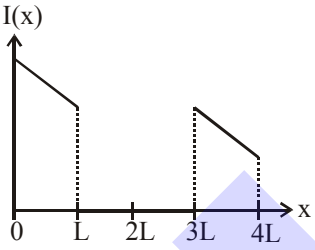
EM0209

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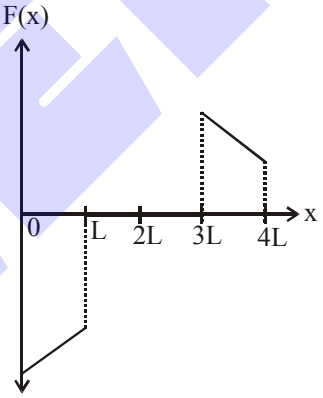
10. A rigid wire loop of square shape having side of length L and resistance R is moving along the x-axis with a constant velocity v_0 in the plane of the paper. At t = 0, the right edge of the loop enters a region of length 3L where there is a uniform magnetic field B_0 into the plane of the paper, as shown in the figure. For sufficiently large v_0, the loop eventually crosses the region. Let x be the location of the right edge of the loop. Let v(x), I(x) and F(x) represent the velocity of the loop, current in the loop, and force on the loop, respectively, as a function of x. Counter-clockwise current is taken as positive. Which of the following schematic plot(s) is(are) correct ? (Ignore gravity) **[JEE Advance-2016]**



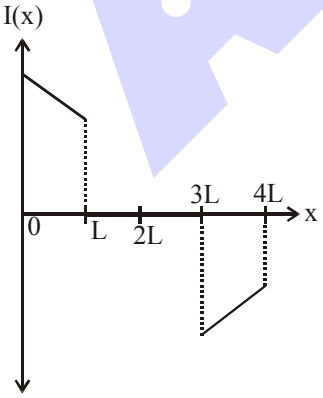
(A)



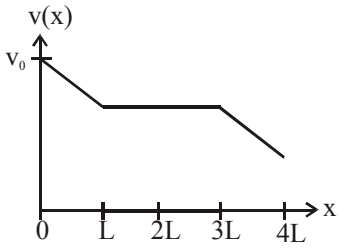
(B)



(C)



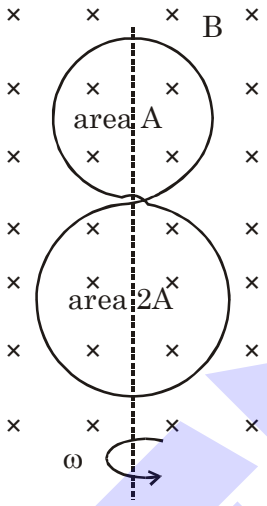
(D)



EM0210

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects of Current EM1 & AC\Eng\02_EM1 & AC\02_Exercises.p65

11. A circular insulated copper wire loop is twisted to form two loops of area A and 2A as shown in the figure. *At the point of crossing the wires remain electrically insulated from each other*. The entire loop lies in the plane (of the paper). A uniform magnetic field $\vec{B}$ points into the plane of the paper. At t = 0, the loop starts rotating about the common diameter as axis with a constant angular velocity ω in the magnetic field. Which of the following options is/are correct? **[JEE Advance-2017]**



(A) The rate of change of the flux is maximum when the plane of the loops is perpendicular to plane of the paper

(B) The net emf induced due to both the loops is proportional to cos ωt

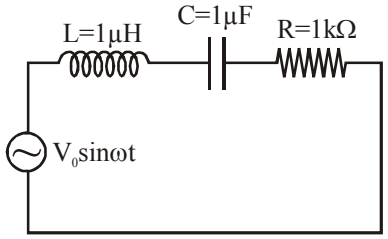
(C) The emf induced in the loop is proportional to the sum of the areas of the two loops

(D) The amplitude of the maximum net emf induced due to both the loops is equal to the amplitude of maximum emf induced in the smaller loop alone

EM0211

12. In the circuit shown, L = 1 μH, C = 1 μF and R = 1 kΩ. They are connected in series with an a.c. source $V = V_0 \sin \omega t$ as shown. Which of the following options is/are correct ?

[JEE Advance-2017]



(A) The frequency at which the current will be in phase with the voltage is independent of R.

(B) At $\omega \sim 0$ the current flowing through the circuit becomes nearly zero

(C) At $\omega >> 10^6$ rad.s^{-1}, the current behaves like a capacitor.

(D) The current will be in phase with the voltage if $\omega = 10^4$ rad.s^{-1}.

EM0212

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

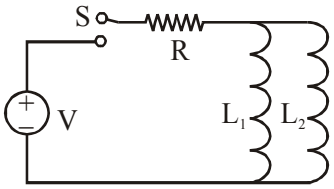
13. A source of constant voltage V is connected to a resistance R and two ideal inductors L_1 and L_2 through a switch S as shown. There is no mutual inductance between the two inductors. The switch S is initially open. At t = 0, the switch is closed and current begins to flow. Which of the following options is/are correct? **[JEE Advance-2017]**

(A) The ratio of the currents through L_1 and L_2 is fixed at all times (t > 0)

(B) After a long time, the current through L_1 will be $\frac{V}{R}\frac{L_2}{L_1+L_2}$

(C) After a long time, the current through L_2 will be $\frac{V}{R}\frac{L_1}{L_1+L_2}$

(D) At t = 0, the current through the resistance R is $\frac{V}{R}$



EM0213

14. The instantaneous voltages at three terminals marked X, Y and Z are given by

$V_X = V_0 \sin \omega t$

$V_Y = V_0 \sin\left(\omega t + \frac{2\pi}{3}\right)$ and $V_Z = V_0 \sin\left(\omega t + \frac{4\pi}{3}\right)$

An ideal voltmeter is configured to read rms value of the potential difference between its terminals. It is connected between points X and Y and then between Y and Z. The reading(s) of the voltmeter will be:- **[JEE Advance-2017]**

(A) $V_{XY}^{rms} = V_0$

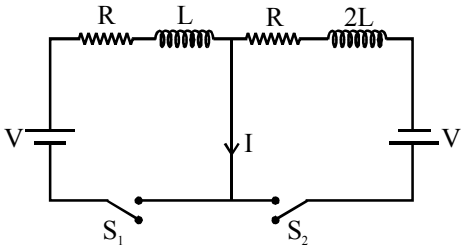
(B) $V_{YZ}^{rms} = V_0\sqrt{\frac{1}{2}}$

(C) Independent of the choice of the two terminals

(D) $V_{XY}^{rms} = V_0\sqrt{\frac{3}{2}}$

EM0214

15. In the figure below, the switches S_1 and S_2 are closed simultaneously at t = 0 and a current starts to flow in the circuit. Both the batteries have the same magnitude of the electromotive force (emf) and the polarities are as indicated in the figure. Ignore mutual inductance between the inductors. The current I in the middle wire reaches its maximum magnitude I_{max} at time $t = \tau$. Which of the following statement(s) is (are) true? **[JEE Advance-2018]**

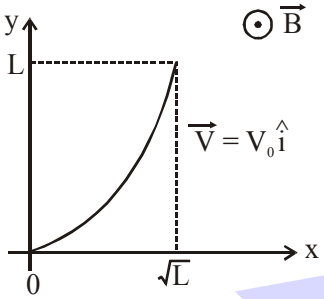


(A) $I_{max} = \frac{V}{2R}$ (B) $I_{max} = \frac{V}{4R}$ (C) $\tau = \frac{L}{R}\ell n2$ (D) $\tau = \frac{2L}{R}\ell n2$

EM0215

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects of Current-EM1 & AC\Eng\02_EM1 & AC\02_Exercises.p65

16. A conducting wire of parabolic shape, initially $y = x^2$, is moving with velocity $\vec{V} = V_0\hat{i}$ in a non-uniform magnetic field $\vec{B} = B_0\left(1+\left(\frac{y}{L}\right)^{\beta}\right)\hat{k}$, as shown in figure. If V_0, B_0, L and β are positive constants and $\Delta\phi$ is the potential difference developed between the ends of the wire, then the correct statement(s) is/are: **[JEE Advance-2019]**



(1) $|\Delta\phi|$ remains the same if the parabolic wire is replaced by a straight wire, y = x initially, of length $\sqrt{2}L$

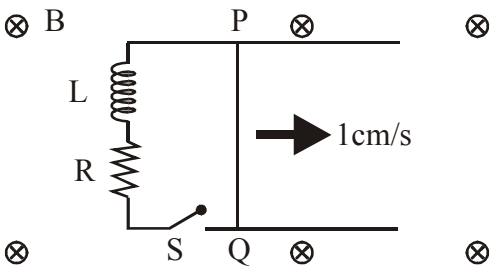
(2) $|\Delta\phi|$ is proportional to the length of the wire projected on the y-axis.

(3) $|\Delta\phi| = \frac{1}{2}B_0V_0L$ for $\beta = 0$

(4) $|\Delta\phi| = \frac{4}{3}B_0V_0L$ for $\beta = 2$

EM0216

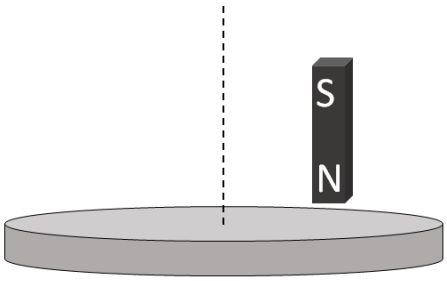
17. A 10 cm long perfectly conducting wire PQ is moving, with a velocity 1cm/s on a pair of horizontal rails of zero resistance. One side of the rails is connected to an inductor L = 1 mH and a resistance R = 1Ω as shown in figure. The horizontal rails, L and R lie in the same plane with a uniform magnetic field B = 1 T perpendicular to the plane. If the key S is closed at certain instant, the current in the circuit after 1 millisecond is $x \times 10^{-3}$A, where the value of x is_____.
[Assume the velocity of wire PQ remains constant (1 cm/s) after key S is closed.
Given : $e^{-1} = 0.37$, where e is base of the natural logarithm] **[JEE Advance-2019]**



EM0217

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

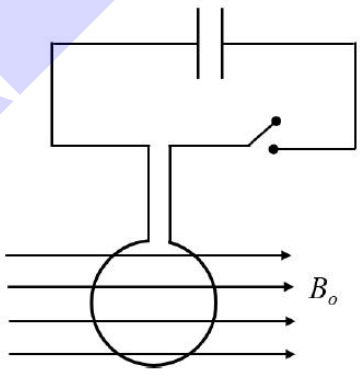
18. A light disc made of aluminium (a nonmagnetic material) is kept horizontally and is free to rotate about its axis as shown in the figure. A strong magnet is held vertically at a point above the disc away from its axis. On revolving the magnet about the axis of the disc, the disc will (figure is schematic and not drawn to scale)- **[JEE Advance-2020]**



(A) rotate in the direction opposite to the direction of magnet's motion
(B) rotate in the same direction as the direction of magnet's motion
(C) not rotate and its temperature will remain unchanged
(D) not rotate but its temperature will slowly rise

EM0252

19. A circular coil of radius R and N turns has negligible resistance. As shown in the schematic figure, its two ends are connected to two wires and it is hanging by those wires with its plane being vertical. The wires are connected to a capacitor with charge Q through a switch. The coil is in a horizontal uniform magnetic field B_0 parallel to the plane of the coil. When the switch is closed, the capacitor gets discharged through the coil in a very short time. By the time the capacitor is discharged fully, magnitude of the angular momentum gained by the coil will be (assume that the discharge time is so short that the coil has hardly rotated during this time)- **[JEE Advance-2020]**

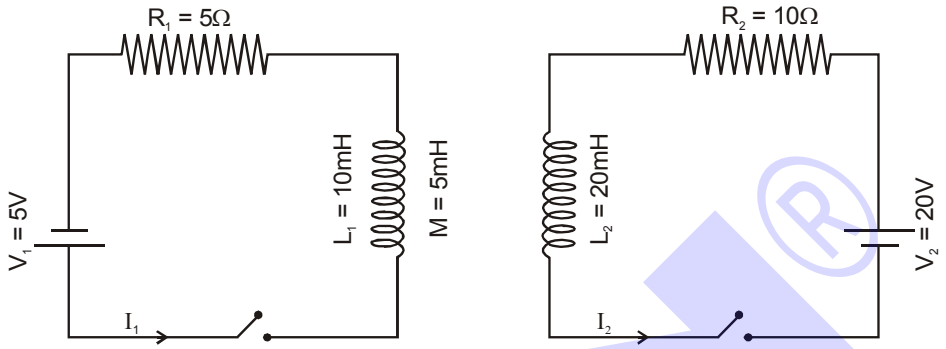


(A) $\frac{\pi}{2}NQB_0R^2$ (B) πNQB_0R^2 (C) $2\pi NQB_0R^2$ (D) $4\pi NQB_0R^2$

EM0253

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects of Current EM1 & AC\Eng\02_EM1 & AC\02_Exercises.p65

20. The inductors of two LR circuits are placed next to each other, as shown in the figure. The values of the self-inductance of the inductors, resistances, mutual-inductance and applied voltages are specified in the given circuit. After both the switches are closed simultaneously, the total work done by the batteries against the induced EMF in the inductors by the time the currents reach their steady state values is_____ mJ. **[JEE Advance-2020]**

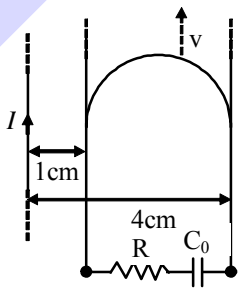


EM0254

21. A long straight wire carries a current, I = 2 ampere. A semi-circular conducting rod is placed beside it on two conducting parallel rails of negligible resistance. Both the rails are parallel to the wire. The wire, the rod and the rails lie in the same horizontal plane, as shown in the figure. Two ends of the semi-circular rod are at distances 1cm and 4 cm from the wire. At time t = 0, the rod starts moving on the rails with a speed v = 3.0 m/s (see the figure).

A resistor R = 1.4 Ω and a capacitor C_0 = 5.0 μF are connected in series between the rails. At time t = 0, C_0 is uncharged. Which of the following statement(s) is(are) correct ?

[$\mu_0 = 4\pi \times 10^{-7}$ SI units. Take ln 2 = 0.7] **[JEE Advance-2021]**



(A) Maximum current through R is 1.2×10^{-6} ampere.

(B) Maximum current through R is 3.8×10^{-6} ampere.

(C) Maximum charge on capacitor C_0 is 8.4×10^{-12} coulomb.

(D) Maximum charge on capacitor C_0 is 2.4×10^{-12} coulomb.

EM0266

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\13-Magnetic Effect of Current+EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

Question Stem for Question Nos. 22 and 23

Question Stem

In a circuit, a metal filament lamp is connected in series with a capacitor of capacitance C μF across a 200 V, 50 Hz supply. The power consumed by the lamp is 500 W while the voltage drop across it is 100 V. Assume that there is no inductive load in the circuit. Take rms values of the voltages. The magnitude of the phase-angle (in degrees) between the current and the supply voltage is φ. Assume, $\pi\sqrt{3} \approx 5$. **[JEE Advance-2021]**

22. The value of C is ____.

EM0267

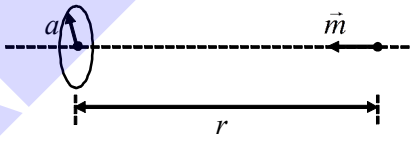
23. The value of φ is ____.

EM0268

Question Stem for Question Nos. 24 and 25

A special metal S conducts electricity without any resistance. A closed wire loop, made of S, does not allow any change in flux through itself by inducing a suitable current to generate a compensating flux. The induced current in the loop cannot decay due to its zero resistance. This current gives rise to a magnetic moment which in turn repels the source of magnetic field or flux. Consider such a loop, of radius a, with its center at the origin. A magnetic dipole of moment m is brought along the axis of this loop from infinity to a point at distance r (>> a) from the center of the loop with its north pole always facing the loop, as shown in the figure below.

The magnitude of magnetic field of a dipole m, at a point on its axis at distance r, is $\frac{\mu_0}{2\pi}\frac{m}{r^3}$, where μ_0 is the permeability of free space. The magnitude of the force between two magnetic dipoles with moments, m_1 and m_2, separated by a distance r on the common axis, with their north poles facing each other, is $\frac{km_1m_2}{r^4}$, where k is a constant of appropriate dimensions. The direction of this force is along the line joining the two dipoles. **[JEE Advance-2021]**



24. When the dipole m is placed at a distance r from the center of the loop (as shown in the figure), the current induced in the loop will be proportional to :

(A) $\frac{m}{r^3}$ (B) $\frac{m^2}{r^2}$ (C) $\frac{m}{r^2}$ (D) $\frac{m^2}{r}$

EM0269

25. The work done in bringing the dipole from infinity to a distance r from the center of the loop by the given process is proportional to :

(A) $\frac{m}{r^5}$ (B) $\frac{m^2}{r^5}$ (C) $\frac{m^2}{r^6}$ (D) $\frac{m^2}{r^7}$

EM0270

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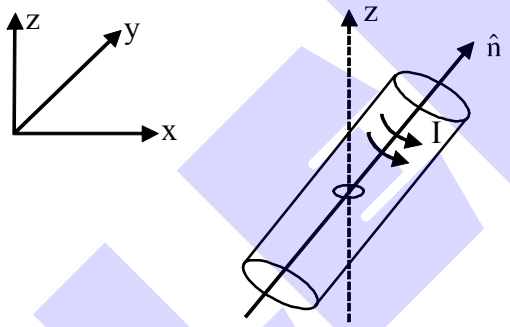
26. Consider an LC circuit, with inductance L = 0.1 H and capacitance C = 10^{-3} F, kept on a plane. The area of the circuit is 1 m^2. It is placed in a constant magnetic field of strength B_0 which is perpendicular to the plane of the circuit. At time t = 0, the magnetic field strength starts increasing linearly as $B = B_0 + \beta t$ with $\beta = 0.04\ Ts^{-1}$. The maximum magnitude of the current in the circuit is____mA.

[JEE Advance-2022]

EM0428

27. A small circular loop of area A and resistance R is fixed on a horizontal xy-plane with the center of the loop always on the axis $\hat{n}$ of a long solenoid. The solenoid has m turns per unit length and carries current I counterclockwise as shown in the figure. The magnetic field due to the solenoid is in direction. List-I gives time dependences of $\hat{n}$ in terms of a constant angular frequency ω. List-II gives the torques experienced by the circular loop at time $t = \frac{\pi}{6\omega}$, Let $\alpha = \frac{A^2\mu_0^2 m^2 I^2 \omega}{2R}$.

[JEE Advance-2022]



	List-I		List-II
(I)	$\frac{1}{\sqrt{2}}\left(\sin\omega t\,\hat{j} + \cos\omega t\,\hat{k}\right)$	(P)	0
(II)	$\frac{1}{\sqrt{2}}\left(\sin\omega t\,\hat{i} + \cos\omega t\,\hat{j}\right)$	(Q)	$-\frac{\alpha}{4}\hat{i}$
(III)	$\frac{1}{\sqrt{2}}\left(\sin\omega t\,\hat{i} + \cos\omega t\,\hat{k}\right)$	(R)	$\frac{3\alpha}{4}\hat{i}$
(IV)	$\frac{1}{\sqrt{2}}\left(\cos\omega t\,\hat{i} + \sin\omega t\,\hat{k}\right)$	(S)	$\frac{\alpha}{4}\hat{j}$
		(T)	$-\frac{3\alpha}{4}\hat{i}$

Which one of the following options is correct?

(A) I → Q, II→P, III→S, IV→T
(B) I→S, II→T, III→Q, IV→P
(C) I→Q, II→P, III→S, IV→R
(D) I→T, II→Q, III→P, IV→R

EM0429

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	C	B	A	D	C	A	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	B	D	D	D	B	B	D	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	C	D	B	A	B	A	D	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	D	A	B	D	B	C	D	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	C	D	B	B	A	A	A	C	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	C	C	A	A	D	B	C	B	C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	A	C	C	D	A	B	A	D	A	B
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	C	A	D	B	A	B	A	B	B	D
Que.	81	82	83	84	85	86				
Ans.	B	A	D	D	A	D				

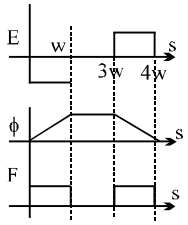
EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	A	D	A	C	A	C	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	D	A	C	A	D	A	C	D	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	B	A	C	B	C	A	D	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	A	D	D	C	B	B	A	A	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	C	C	B	D	A	A	A	B	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	D	C	C	B	A	B,C	A,C	A,B	B	A,B,C,D
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	A,C	A,C,D	D	B,D	B,D	A	B	A,C,D	A,C	A,C
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	A,B	B,C	C or A,C	A,B,C	A,C,D	C	C	B	B	D
Que.	81	82	83	84	85	86	87	88		
Ans.	B	C	D	C	B	D	C	(A)-P; (B)-PQS; (C)-QS; (D)-QRS		
Que.	89				90					
Ans.	(A)-Q; (B)-R,S; (C)-S; (D)-P,Q,R				(A)-R,S,T; (B)-Q,R,S,T; (C)-P,Q; (D)-Q,R,S,T					

node06\B0BC-BD\Work\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effects\Current\EMI & AC\Eng\02_EMI & AC\02_Exercises.p65

EXERCISE # S

1. (a) 0; (b) $B_H S$; (c) $-B_V S$ **2.** 0.75 T **3.** 6.5 V **4.** $\lambda V_y B_0$ **5.** 2 N

6. 

7. $\frac{mgR}{B^2\ell^2}$ **8.** 0.21 V **9.** 720 volt, 864 newton meter **10.** 6

11. 5 **12.** (i) 85.22 Tm^2; (ii) 56.8 V; (iii) linearly

13. (a) zero, (b) vB(bc), positive at c, (c) zero (d) vB(bc), positive at a

14. (a) $\frac{RF - vd^2B^2}{mR}$; (b) $\frac{RF}{d^2B^2}$ **15.** $I_{ind} = \alpha/r$, where $\alpha = \frac{1}{2}\mu_0 \ell vI/\pi R$

16. $V = 1\ ms^{-1}$, $R_1 = 0.47\ \Omega$, $R_2 = 0.30\ \Omega$ **17.** $\frac{1}{3}A$ **18.** (i) 2.4×10^{-5} V (ii) from c to b

19. $\frac{erk}{2m}$ directed along tangent to the circle of radius r, whose centre lies on the axis of cylinder.

20. $\frac{l}{2}\frac{dB}{dt}\sqrt{R^2 - \frac{l^2}{4}}$ **21.** (a) 8×10^{-21} N (b) 0.3 V/m **22.** 6 **23.** $\frac{\mu_0 i a^2 \pi}{2Rb}$ **24.** I^{-1}

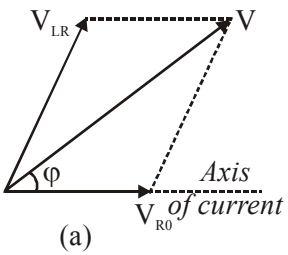
25. 30 Wb. **26.** $\varepsilon = 1.7 \times 10^{-5}$V **27.** 1.25 A **28.** 7 **29.** $kMT^2/(R)$

30. 0.8 **31.** $\frac{LE^2}{2R_1^2}$ **32.** $\frac{e^2}{e^2 - 1}$ **33.** $\frac{EL}{eR^2}$ **34.** 5

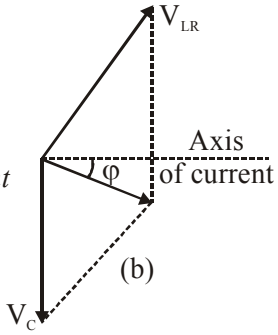
35. (i) $i_1 = i_2 = 10/3$ A, (ii) $i_1 = 50/11$ A ; $i_2 = 30/11$ A, (iii) $i_1 = 0$, $i_2 = 20/11$ A, (iv) $i_1 = i_2 = 0$

36. 42 + 20t volt **37.** $\frac{1}{15}A, \frac{1}{10}A$ **38.** $q = Q_0 \sin\left(\sqrt{\frac{1}{LC}}t + \frac{\pi}{2}\right)$

39. (a)10^4A/s; (b) 0; (c) 2A; (d) $100\sqrt{3}\ \mu C$ **40.**



(a)



(b)

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41. 0.08 H, 17.28 W **42.** 2A, 400W **43.** 0.2 mH, $\frac{1}{32}$ μF, 8×10^5 rad/s

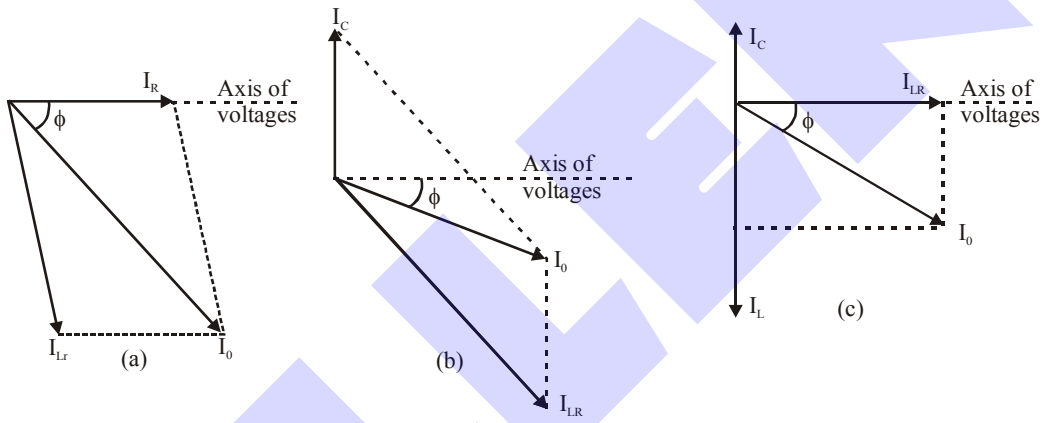
44. $\frac{20}{\pi^2} \cong 2H$ **45.** 20 V **46.** 20 A, π/4,

47. (a) 600 kW (b) 1400 kW (c) 440 - 7000 V

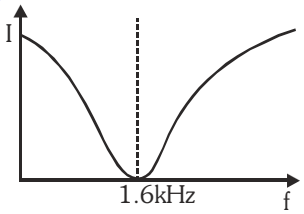
48. (a) I leads V by $\frac{\pi}{3}$; (b) 1kΩ , 500Ω, $\frac{500}{\sqrt{3}}\Omega$; (c) 0.5, 2.5W; (d) Circuit must contains R as $\phi \neq \frac{\pi}{2}$ and as ϕ is negative so C must be their, (L may exist but $X_C > X_L$)

49. $\frac{5\sqrt{7}}{2}$A **50.** 4

51.



52. 4 Ω, 1.5 A **53.** 1.592 kHz, **54.** 120 V



55. R = 240 Ω, L = 1.5 H **56.** 77Ω, 97.6Ω, 7.7 V, 9.76 V

57. $Q = \sqrt{\frac{\eta^2 - 1}{(n-1)^2} - \frac{1}{4}}$ **58.** 0.05 **59.** $\frac{FA\omega}{2}\sin\varphi$

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	1	3	1	2	3	1	1	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	4	2	2	3	2	2	3	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	2	2	By NTA (200) By Allen (2000)	2	1	By NTA (400) By Allen (282.84)	3	3
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	1	4	6	4	1	3	3	1	80	3
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	Drop	3	1	2	A	4	D	B	C	3
Que.	51	52	53							
Ans.	12	250	C							

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	A	B	C,D	A,D	A,C	B,D	8	C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,D	A,B	A,B,C	C,D	B,D	1,2,4	0.63	B	B	55.00
Que.	21	22	23	24	25	26	27			
Ans.	A,C	100.00	60.00	A	C	3.98 - 4.02	Dropped			

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